EEG-Based Motor Imagery BCI Algorithms and Technologies: A Review

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***Abstract*—Brain-computer interfaces (BCIs) have emerged as transformative technologies that enable direct communication between the brain and external devices. Among various BCI paradigms, EEG-based motor imagery (MI) has gained prominence due to its simplicity, non-invasiveness, and potential to restore motor function and facilitate rehabilitation for patients with motor impairments. This paper presents a comprehensive review of the most practical processing algorithms developed over the past decade for decoding brain sensorimotor cortex signals. Specifically, this paper discusses the integration of artificial intelligence (AI)-based algorithms, particularly machine learning and deep learning techniques, and their contributions to improving the performance and efficiency of MI-BCI systems in detail. Furthermore, the paper reviews state-of-the-art hardware platforms and emerging converging technologies—including system-on-chip (SoC) architectures, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), wearable devices, the Internet of Things (IoT), and augmented/virtual reality (AR/VR)—and discusses their integration with advanced signal processing algorithms to enable next-generation MI-BCI systems. By highlighting current achievements of EEG-based MI-BCI technology and predicting future research directions that could further enhance real-time capabilities, this paper aims to provide valuable insights for researchers and practitioners, fostering innovation in high-performance EEG-based MI-BCI systems.**

***Index Terms*—Brain-computer interface, BCI, motor imagery, electroencephalography, EEG, signal processing, artificial intelligence, machine learning, deep learning, transformers, neural networks, convolutional neural networks, CNN, SoC.**

I. Introduction

THE history of brain-computer interfaces (BCIs) spans nearly a century, beginning with Hans Berger's pioneering work in 1924 when he first recorded human brain activity using electroencephalography (EEG), which provided the first evidence that neural activity associated with mental processes can be measured through electrical brain signals [1]–[3]. BCIs represent a transformative communication paradigm that establishes a direct, neuromuscular-independent pathway between the central nervous system and external computational devices [4]–[7]. By definition, a BCI is a hardware-software system that acquires neural activity, processes it in real-time, and translates it into discrete control commands for assistive tools, neuroprosthetics, or digital interfaces [8]–[10]. Among the various brain-signal acquisition modalities, EEG records scalp electrical potentials arising predominantly from the summed postsynaptic activity of neuronal populations and remains one of the most widely used non-invasive approaches for BCI applications because of its high temporal resolution, portability, relatively low cost, and ease of use [11]–[14].

Early experiments demonstrating that animals could control devices through neural activity paved the way for significant developments in neuroprosthetics [8]–[10]. These studies ultimately aimed to restore lost motor and communication functions. The mid-1990s saw the advent of the first neuroprosthetic devices for humans, demonstrating the potential of BCIs to enhance the quality of life for individuals with severe disabilities [15]–[18]. Today, as shown in Fig. 1, BCIs are recognized not only for their profound medical applications but also for their expanding non-medical utility and potential to augment human capabilities, making them a critical area of research spanning neuroscience and engineering [19]–[21].

Within the medical domain, EEG-based BCIs are pivotal for motor imagery (MI) tasks, crucial for rehabilitation and control, as well as for disease detection and sleep analysis. Concurrently, non-medical applications are rapidly emerging, including emotion recognition, fatigue detection for safety and performance monitoring, and interactive gaming [22], [23]. As technology continues to evolve, the implications of BCIs extend into ethical and philosophical domains, challenging our understanding of consciousness and human-machine interaction [24]–[27].

EEG-based motor imagery BCIs (MI-BCIs) enable individuals to control external devices by imagining limb movements, without any physical action [28], [29]. This non-invasive approach offers significant advantages, including its portability, relatively low cost, and ease of use compared with invasive BCI approaches [30], [31]. These benefits make MI-BCIs highly accessible for widespread research and practical applications, particularly in rehabilitation and assistive technologies for individuals with motor impairments [32], [33].

This paper provides a comprehensive review of EEG-based MI-BCI systems, with a specific focus on system architecture and the complete signal-processing pipeline, including benchmark datasets, software and hardware implementations, and emerging enabling technologies for next-generation wearable and edge MI-BCI systems. Unlike broader BCI surveys or reviews that focus primarily on algorithms, rehabilitation, or in-

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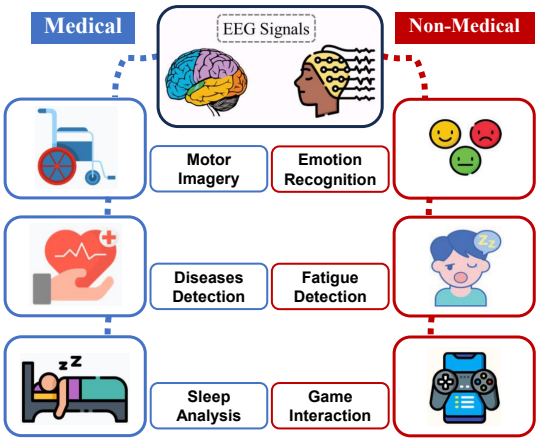

Fig. 1. Various medical and non-medical applications of EEG-based BCI modality.

dividual technologies, this work jointly examines classical and AI-based signal-processing methods, hardware realization, and converging technologies within a unified MI-BCI framework. By providing a foundational roadmap for researchers and engineers, this paper addresses the need for robust, "plug-and-play" assistive MI-BCI technologies capable of transitioning from laboratory settings to real-world clinical environments.

The remainder of this paper is organized as follows: Section II details our review methodology, outlining the literature review approach and research questions. Section III explores various BCI modalities, contrasting electrical, magnetic, and metabolic methods before focusing on EEG-based BCIs and their applications, specifically highlighting the advantages of motor imagery. Section IV delves into the fundamental EEG-based MI-BCI system functional blocks. Subsequently, Section V presents a comprehensive analysis of MI-BCI signal processing algorithms, covering both software-based (classic and AI algorithms) and hardware-based approaches. Section VI examines converging technologies for the next generation of MI-BCI Systems. Building upon these foundations, Section VII proposes a sample next-generation EEG-based MI-BCI system, detailing its key subsystems and an AI-based signal processing architecture. Finally, Section VIII offers our discussions, answering the research questions, identifying key opportunities and cross-disciplinary challenges, and providing concluding remarks on the future of the field. Section IX concludes the paper.

II. Review Methodology

To investigate the dominant BCI applications, the signal processing techniques used for EEG-based MI-BCI systems over the past decade, and emerging technologies for future MI-BCI systems, the following research questions (RQs) are defined.

RQ1. Which BCI application currently has the greatest practical impact and the strongest potential for future development and improving the quality of life? What is the role of EEG-based MI-BCI systems within this landscape?

RQ2. What are the most frequently used signal processing techniques and datasets for EEG-based MI-BCI systems in the recent decade? Which processing approaches have demonstrated superior performance in recent studies and have the most potential to be used in future systems?

RQ3. What are the challenges of EEG-based MI-BCI systems, and what are the promising candidate solutions?

RQ4. How can various emerging technologies be brought together to create the next generation of EEG-based MI-BCI systems?

To answer the above questions, this study employs a structured review methodology to minimize selection bias and ensure a reproducible, transparent synthesis of technical data, as discussed in the following sections.

A. Review Design and Search Strategy

The review was conducted in two stages. First, to address the preceding RQs, the broader EEG-based BCI literature was mapped, and a statistical analysis of article titles and abstracts was performed to identify major application areas and emerging technological trends in the BCI field. As discussed in detail in Section VIII, this analysis indicates that EEG-based MI-BCI systems for medical and rehabilitation applications represent a major area of research within the field. Relevant studies were retrieved from IEEE Xplore, PubMed, ScienceOpen, Semantic Scholar, Scopus, Web of Science, and Google Scholar. The literature searches were conducted in July 2026, with the publication period restricted to January 1, 2014, through July 1, 2026. The complete search strings and the corresponding numbers of retrieved records are reported in Table I. The same conceptual search strings were applied across all databases, with minor syntax adaptations according to the search interface supported by each database. Frequently used technical terms and their abbreviations are listed in Table II.

Second, the review focused specifically on MI-BCI studies, systematically searched across IEEE Xplore, PubMed, ScienceOpen, Semantic Scholar, Scopus, Web of Science, and Google Scholar using predefined EEG, BCI, MI, and technology-specific keywords. Influential studies from the recent decade, including recent works proposing novel methodologies, were analyzed with emphasis on signal-processing algorithms, benchmark datasets, hardware implementations, and enabling technologies. The resulting findings are summarized and discussed in the following sections, while a comprehensive classification of commonly used algorithms is presented in Fig. 25.

B. Eligibility Criteria

The eligibility criteria for selecting works at both stages included studies published from January 1, 2014, to July 1, 2026, focusing on journal articles, conference proceedings, and dataset reports. Moreover, studies were eligible for the focused MI-BCI synthesis when they: (1) reported original research in a peer-reviewed journal or conference proceeding, or described a benchmark dataset relevant to MI-BCI research; (2) investigated EEG-based BCI with an explicit motor-imagery component; and (3) provided sufficient methodological or quantitative information concerning signal

TABLE I
SEARCH STRINGS AND NUMBERS OF RECORDS RETRIEVED FROM THE SELECTED DATABASES FOR EEG-BASED BCI AND RELATED TECHNOLOGIES. SEARCHES WERE CONDUCTED IN JULY 2026 AND RESTRICTED TO PUBLICATIONS FROM JANUARY 1, 2014, TO JULY 1, 2026.

Search Strings / Search Engine	IEEE Xplore	PubMed	ScienceOPEN	Semantic Scholar	Scopus	Web of Science	Google Scholar
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface")	11,751	6,770	3,610	11,900	4,916	8,665	22,600
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("motor imagery" OR "motor-imagery" OR MI)	3,267	2,062	1,235	7,190	2,301	3,862	17,900
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("machine learning" OR ML OR "deep learning" OR DL OR "neural network" OR "artificial intelligence" OR AI)	5,230	2,961	1,547	4,040	1,101	2,449	18,100
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND (wearable OR wireless)	806	349	287	3,710	1,318	776	17,700
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("system-on-chip" OR "system on a chip" OR SoC OR "hardware accelerator" OR "hardware acceleration")	398	696	234	3,400	44	106	6,190
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("Internet of Things" OR IoT)	284	29	28	301	376	99	11,200
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("energy harvesting" OR "power harvesting" OR "energy scavenging" OR EH OR "low power" OR LP)	714	211	179	849	119	723	2,120
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("edge computing" OR "edge processing")	148	49	35	863	1,306	175	17,200
(EEG OR electroencephalographic OR electroencephalography) AND (BCI OR "brain-computer interface") AND ("virtual reality" OR VR)	413	221	186	3,280	940	659	17,300

TABLE II
LIST OF FREQUENTLY USED ABBREVIATIONS IN THIS PAPER.

Full Term	Abbreviation	Full Term	Abbreviation	Full Term	Abbreviation	Full Term	Abbreviation
Motor imagery-based BCI	MI-BCI	Brain-computer interface	BCI	Artificial intelligence	AI	Electroencephalography	EEG
Machine learning	ML	Artificial neural network	ANN	Deep learning	DL	Generative AI	GenAI
Classification accuracy	CA	Convolutional neural network	CNN	Feedforward neural network	FFNN	Back propagation neural network	BPNN
Probabilistic neural network	PNN	Euclidean Space	EUS	System-on-Chip	SoC	Band-pass filtering	BPF
Hybrid	HYB	Deep belief network	DBN	Transformer	TRF	Common spatial pattern	CSP
Support vector machine	SVM	Linear discriminant analysis	LDA	Neuro-fuzzy system	NFS	Hardware accelerator	HA
Internet of Things	IoT	Mutual information	MIN	Compressive sensing	CPS	Energy harvesting	EH
Autoencoder	AEC	Fuzzy integral	FZI	Edge processing	EP	Transfer learning	TL
Virtual reality	VR	Digital signal processing	DSP	Naive bayes	NVB	long short-term memory	LSTM

preprocessing, feature extraction or selection, classification, datasets, experimental validation, or hardware implementation.

Studies were excluded from the focused synthesis when they: (1) investigated exclusively non-EEG modalities such as ECoG, MEG, or fMRI; (2) investigated EEG paradigms unrelated to motor imagery, such as exclusively SSVEP- or P300-based systems, unless they were used only for contextual comparison; (3) addressed EEG applications without a BCI component; (4) were reviews, editorials, news articles, commercial webpages, or other non-primary sources used only for background information; (5) duplicated another retrieved publication; or (6) did not provide sufficient methodological or outcome information for extraction. Studies outside the focused MI-BCI scope were not included in the quantitative synthesis even when they were cited elsewhere in the manuscript for general technological background.

C. Study Screening and Deduplication

Duplicate records were first identified using DOI, PubMed ID, and exact-title matching, followed by manual comparison of titles, authors, publication years, and journal/conference information to identify remaining duplicates. After deduplication, 1,247 unique records remained. Titles and abstracts were then screened against the eligibility criteria, resulting in the exclusion of 349 records. The full texts of the remaining 898 publications were assessed for eligibility, and 152 publications were excluded because of non-MI paradigms, non-EEG modalities, insufficient methodological information, duplicate or secondary reports, irrelevant applications, or other scope-related reasons. The remaining 746 studies constituted the relevant literature corpus. From these, 324 studies were selected for detailed analysis and inclusion in this review based on their relevance to the specific scope, research questions, and

objectives of the paper.

D. Quality Assessment

The quality of the studies was evaluated based on: (1) adequacy of the description of participants or dataset provenance; (2) reproducibility of EEG preprocessing and feature-processing procedures; (3) completeness of model architecture and training details; (4) appropriateness and transparency of the validation protocol, including separation between training and test data and specification of subject-dependent or subject-independent evaluation; (5) completeness of performance reporting, including relevant accuracy, F1-score, Cohen's κ, variability, or other applicable metrics; and (6) comparison with appropriate baselines or benchmark methods. For hardware-oriented studies, additional consideration was given to whether the implementation platform and validation conditions were adequately described and whether relevant quantities such as power consumption, latency, memory/resource utilization, throughput, or silicon area were reported when applicable.

E. Data Extraction and Synthesis

A standardized data-extraction form was used for all studies included in the focused synthesis. The extracted information included publication year, EEG dataset or participant cohort, number of subjects, number of channels, sampling rate, number and type of MI classes, preprocessing procedures, feature-extraction and feature-selection methods, classification algorithm, validation protocol, subject-dependent or subject-independent evaluation setting, and reported performance metrics. For hardware implementations, the hardware platform and available implementation metrics, including resource utilization, power consumption, area, latency, inference time, memory requirements, and computational efficiency, were additionally extracted.

The percentages reported in Figs. 30 and 32 were calculated as $100 \times n_i/N$, where n_i is the number of studies assigned to category i, and N is the total number of studies included in the corresponding analysis. The denominator N for each analysis is explicitly reported in the respective figure caption. Publication trends in Figs. 31 and 34 were derived from annual Scopus search-result counts using the search strategy described in Section II. The plotted lines simply connect the annual publication counts to visualize temporal trends; no statistical regression, smoothing, or extrapolation was applied. Fig. 33 represents a qualitative synthesis of the challenges and corresponding solutions identified across the reviewed literature, with supporting references provided in the accompanying discussion.

III. BCI MODALITIES

BCI systems are fundamentally built upon a diverse array of neuroimaging modalities, each employing different techniques to capture specific aspects of brain activity. These modalities differ considerably in their underlying physiological mechanisms for signal acquisition, leading to a spectrum of performance trade-offs in terms of resolution, invasiveness, cost, and practicality [34]–[36]. The primary categories, as depicted in Fig. 2, include magnetic modalities, which detect the faint magnetic fields produced by neuronal activity; electric modalities, which measure electrical potentials on the scalp or within the brain; and metabolic modalities, which track changes in blood flow or oxygenation as indicators of neural function [7], [37].

Magnetic-based approaches, such as magnetoencephalography (MEG), excel at capturing the magnetic fields generated by neuronal currents, providing exceptional temporal resolution crucial for understanding the precise timing of brain processes. This makes MEG invaluable for fundamental neuroscience research, allowing scientists to observe the rapid dynamics of neural oscillations and event-related fields [38], [39]. However, the routine application of MEG in practical settings is severely hindered by its requirement for large, prohibitively expensive, and immobile instrumentation, typically housed in specialized magnetically shielded rooms. Consequently, its use is largely confined to controlled laboratory environments rather than routine clinical or consumer applications [40]–[42].

Electric-based BCI modalities offer a broad spectrum of approaches for capturing neural activity, ranging from highly precise invasive techniques to more accessible non-invasive methods. Invasive techniques, such as electrocorticography (ECoG) and local field potentials (LFP), involve direct access to the cortical surface or deeper brain structures, providing neural recordings with high signal-to-noise ratio (SNR) and superior spatial resolution [43], [44]. This makes them invaluable for applications like advanced seizure detection and prediction in epilepsy patients or for sophisticated motor control prosthetics in individuals with severe paralysis, where precise neural decoding is paramount. In contrast, non-invasive EEG-based BCIs record electrical activity from the scalp without requiring surgical implantation [3], [45], [46]. While these methods are valued for their ease of setup, minimal risk, and affordability, enabling widespread use in clinical diagnostics and consumer-level BCI applications, they inherently sacrifice spatial resolution due to signal distortion and attenuation through the skull and tissues.

Metabolically based BCIs, most notably near-infrared spectroscopy (NIRS) and functional magnetic resonance imaging (fMRI), infer neural activity indirectly by tracking hemodynamic responses, the localized changes in blood flow and oxygenation that occur in response to neuronal activation [47], [48]. By leveraging these metabolic proxies, these modalities provide valuable spatial accuracy, allowing researchers and clinicians to precisely map cognitive functions to specific cortical regions [47], [49]. This makes them well-suited for applications such as pre-surgical mapping for tumor removal, where identifying critical functional areas is vital, or in longitudinal studies of neuroplasticity and cognitive recovery. However, these benefits are frequently counterbalanced by inherent limitations: fMRI, while highly precise, requires large-scale, stationary, and expensive scanners that restrict participant mobility, while NIRS, though more portable and cost-effective, suffers from slower temporal resolution due to the inherent slow dynamics of the vascular response, which creates a significant latency compared to direct electrical

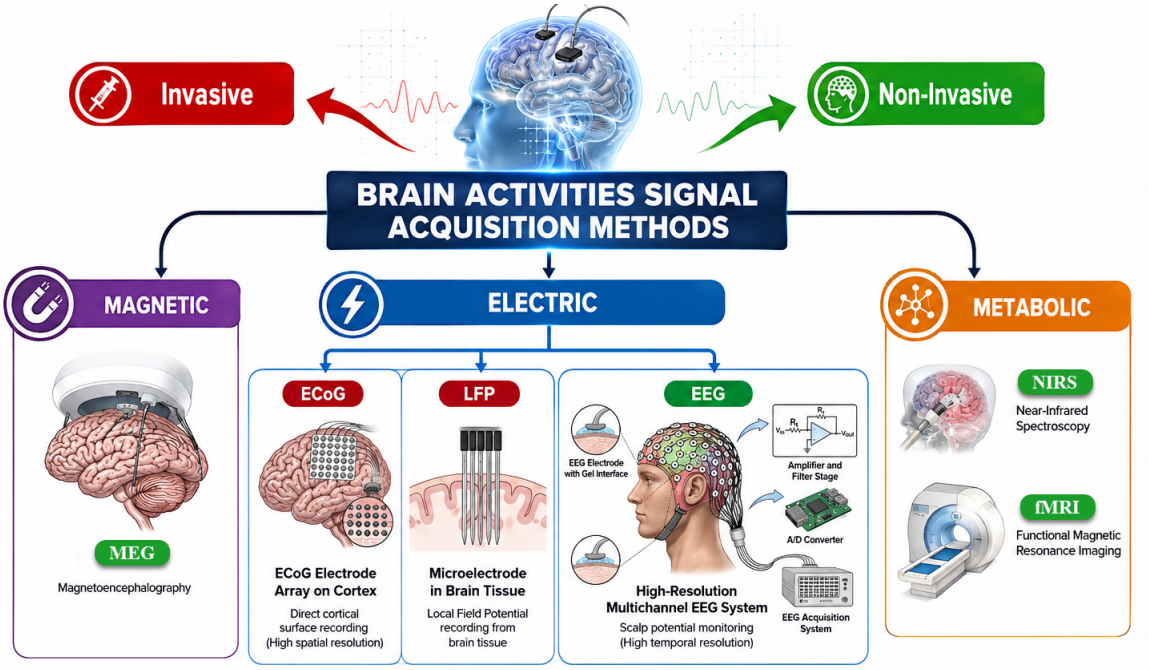

Fig. 2. Overview of brain activity signal acquisition modalities for BCI systems. The modalities are grouped into magnetic (MEG), electric (ECoG, LFP, and EEG), and metabolic (NIRS and fMRI) approaches, highlighting their invasive and non-invasive implementations and representative sensing hardware.

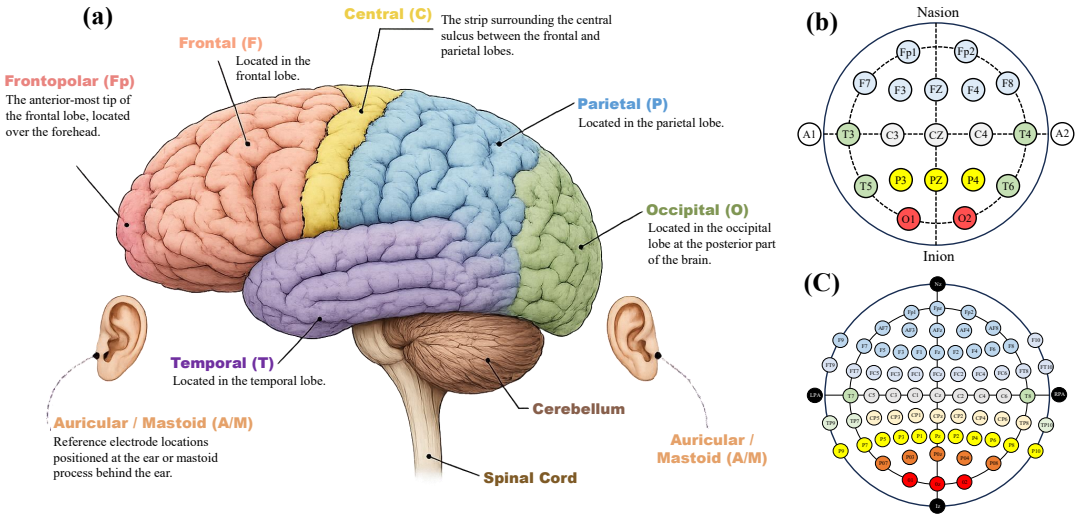

Fig. 3. Brain region labeling and the international standard EEG electrode placement systems. (a) Lateral brain view divided into the six main cortical regions referenced by electrode nomenclature: frontopolar (Fp), frontal (F), central (C), parietal (P), occipital (O), and temporal (T), with the cerebellum and spinal cord shown for anatomical context. (b) 10-20 electrode placement system (21 electrodes), with electrode labels color-coded by region and referenced to the nasion and inion. A1/A2 denote auricular (ear) reference electrodes, which are not cortical sites. (c) 10-10 electrode placement system (extended array of up to 81 electrodes), providing finer spatial resolution between the standard 10-20 sites.

measurements [48], [50].

Recent technological advances have encouraged the development of hybrid (HYB) BCIs that combine multiple modalities to improve performance [51]. For example, by integrating EEG's high temporal resolution with NIRS's superior spatial localization of hemodynamic responses, these HYB systems can provide more robust interfaces. In particular, a HYB BCI may enable more reliable control of a prosthetic limb by using EEG for rapid detection of motor intention and NIRS to complement the activation of specific cortical regions, thereby enhancing responsiveness [52], [53]. In individuals with communication impairments, this combination can support more effective assistive technologies by enabling faster symbol selection through the complementary information (cross-validation) from electrical and metabolic brain signals, thus increasing reliability. In addition, several studies have reported the use of other sensors alongside EEG devices, including electromyography (EMG) [54]–[57], functional electrical stimulation (FES) devices [58], NIRS tools [59], and MRI devices [60].

EEG-based BCIs have gained widespread adoption across diverse domains, including healthcare, rehabilitation, enter-

tainment, education, and automotive systems [61]–[63]. Their medical applications span a broad spectrum, supporting tasks such as emotion recognition [64], [65], detection of seizure and epilepsy disorders [66]–[72], assessment of neurodegenerative diseases such as Parkinson's and Alzheimer's [73]–[75], tumor evaluation [76], and identification of sleep-related abnormalities [77]–[79]. The suitability and widespread choice of EEG for MI paradigms stem from several key advantages: its excellent temporal resolution allows for the detection of neural activity within milliseconds, crucial for capturing the real-time electrical signals generated during the mental rehearsal of movements [80]; its relative non-invasiveness, portability, and cost-effectiveness compared to other neuroimaging techniques make it accessible for diverse applications; and specific patterns of brain activity, such as changes in sensorimotor rhythms, are reliably detectable and provide a clear neural correlate for decoding user intentions. These characteristics collectively make EEG a well-suited and frequently chosen modality for developing and deploying MI-BCI systems [62], [77].

The anatomical brain regions commonly referenced in EEG electrode nomenclature are illustrated in Fig. 3 (a). The cerebral cortex is divided into six major regions: frontopolar (Fp), frontal (F), central (C), parietal (P), occipital (O), and temporal (T), corresponding approximately to the cortical regions from which EEG signals primarily originate. The central region encompasses the sensorimotor cortex along the central sulcus, which plays a crucial role in MI-based BCI applications. In addition, the auricular/mastoid (A/M) locations are shown as extracerebral reference sites positioned at the ear and mastoid process, outside the cerebral cortex. This regional labeling provides a neuroanatomical framework for understanding EEG electrode placement and the cortical origins of recorded neural activity [34], [81], [82].

EEG systems measure potential differences generated primarily by synchronized postsynaptic cortical activity through electrodes placed on the scalp, typically referenced to a common electrode and recorded by an amplifier at fixed sampling rates [29], [83]. Two international electrode placement configurations are the 10-20 system and the 10-10 system, as depicted in Fig. 3 (b) and Fig. 3 (c), respectively. The 10–20 system standardizes electrode placement using proportional distances between anatomical landmarks, enabling consistent labeling of Fp, F, C, P, O, and T regions, while auricular (A) sites located near the ears serve as reference points. For denser recordings, the 10–10 configuration refines the 10–20 layout by adding more electrodes at intermediate standardized intervals, yielding finer spatial sampling across the same labeled regions. Note that compound labels (e.g., AF, FC, etc.) denote electrode positions intermediate between two adjacent regions. For example, "FC3" lies between frontal and central sites. Nz and Iz mark the nasion and inion; LPA/RPA (or A1/A2) denote left/right auricular reference points [33], [84].

For MI-BCI applications, electrode count is a trade-off between information richness and system complexity: fewer electrodes (e.g., 4–8) reduce cost, setup time, and user burden, but they may miss weaker or spatially distributed sensorimotor patterns, whereas more electrodes (e.g., 16–32) improve

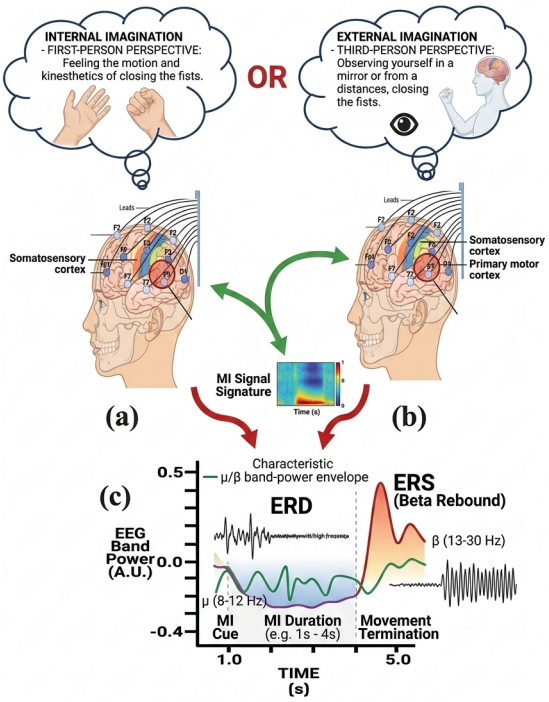

Fig. 4. The MI process in the brain. The first (internal) (a) somatosensory or third (external) (b) observational perspective imagination of opening/closing the fists activates the primary sensorimotor cortex. During imagined movement (c), the specific cortical oscillation rhythms (μ, 8–12 Hz and β, 13–30 Hz) are suppressed, known as Event-Related Desynchronization (ERD). Upon movement termination, the termination of MI elicits a distinct β-band amplitude increase, known as Event-Related Synchronization (ERS) or β-rebound.

spatial coverage and decoding robustness, especially across inter-subject variability, at the cost of longer preparation, greater computational complexity, and increased susceptibility to noisy channels [29], [83]. This is particularly relevant for wearable devices and deployments in uncontrolled environments with limited computational power, where reducing data dimensionality is essential, and classification should rely on as few inputs as possible. Accordingly, beyond traditional feature reduction and selection methods, many studies focus on specific EEG channels, usually chosen from the central cortical region, which is consistent with neuroscientific evidence on MI. In practice, MI-focused montages prioritize the sensorimotor cortex, targeting the μ and β rhythms over the central area, so electrode selections are commonly centered on C-region positions and often include nearby F and P electrodes, with practical setups frequently using around 16 electrodes, often following the international 10–20 electrode placement standard, to balance performance and usability [1], [2], [29].

In the next section, the theoretical fundamentals of the MI paradigm are briefly reviewed, and the functional blocks of practical EEG-based MI-BCI systems are introduced.

IV. EEG-BASED MI-BCI SYSTEMS

Among the various BCI paradigms, MI has become a widely adopted paradigm for neuroprosthetics and rehabilitation due to its unique ability to engage the sensory-motor cortex and activate neural pathways similar to those utilized in actual

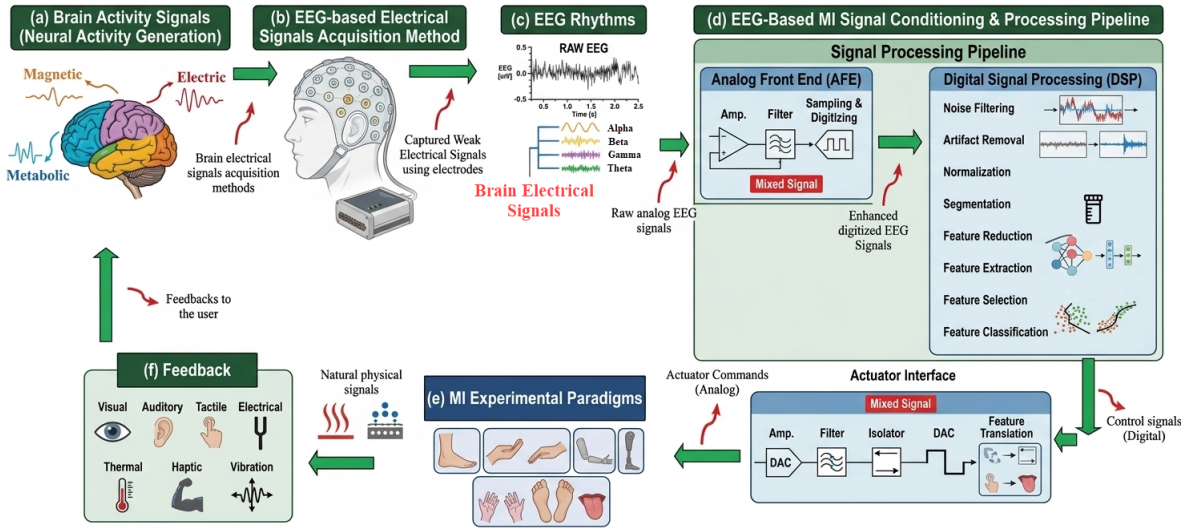

Fig. 5. The EEG-based MI-BCI life cycle overview. (a) Brain electrical, magnetic, and metabolic signals are generated by neural activity. (b) EEG-based electrical signal acquisition method. (c) Five frequency bands of EEG signals, known as EEG rhythms. (d) MI processing pipeline: analog front end (AFE), digital signal processing (DSP), and actuator interface. (e) Conventional MI experimental paradigms. (f) Actuator to brain feedback mechanisms: Vibration, Tactile, Thermal, Electrical, Haptic, Auditory, and Visual.

motor execution [15]–[19], [21]. By allowing users to control external devices through imagined movements, MI-based BCIs provide a vital non-invasive means for individuals with severe disabilities to regain autonomy and interact with their environment despite severe motor impairments [85]–[88]. The primary advantage of these applications is the ability to harness neural signals without requiring residual muscle activity, which is crucial for patients with conditions such as spinal cord injuries [89], [90]. However, transitioning MI-BCI from controlled laboratory settings to real-world clinical use presents significant engineering hurdles, including signal non-stationarity and subject variability. Overcoming these barriers necessitates a symbiotic optimization of both hardware acquisition layers and software classification pipelines to ensure the low latency and high accuracy required for robust, user-friendly assistive technologies [91], [92].

MI refers to the mental simulation of a movement without any physical execution, and it can be carried out from either a first-person (internal) perspective, where the individual feels as if they are performing the movement, or a third-person (external) perspective, in which the individual imagines watching themselves perform the action, as illustrated in Fig. 4 (a) and Fig. 4 (b), respectively. Extensive research shows that imagined and executed movements engage functionally similar neural mechanisms and activate the same primary sensorimotor regions [22], [30]. When a person imagines opening or closing their fists, the brain generates characteristic changes in the μ (8–12 Hz) and β (13–30 Hz) rhythms originating from the primary sensorimotor cortex of the brain. These changes appear as event-related desynchronization (ERD), a suppression in rhythm amplitude that occurs before and during imagined movement, and event-related synchronization (ERS), an increase in β-band amplitude associated with movement termination or cortical idling, as shown in Fig. 4 (c). The MI-induced ERD/ERS patterns provide the fundamental neural signatures used by EEG-based MI BCIs to decode user intent [82], [93], [94].

The block diagram of an MI-BCI system represents a multi-stage architecture designed to decode neural signals and translate them into external commands [6], [95], as conceptually illustrated in Fig. 5. Each interconnected stage plays a critical role in this complex process, from the initial detection of brain activity to the final execution of an action. These stages provide the interface between a user's thoughts and the operation of external devices, offering a powerful means of communication and control for individuals with severe motor impairments. The subsequent paragraphs will delve into the specifics of each of these crucial components.

At the beginning of the MI process, neural activity gives rise to electrical, magnetic, and hemodynamic signals that reflect the underlying neural processes (Fig. 5 (a)). In EEG-based MI-BCI systems, the focus is on capturing the electrical signals. The EEG BCI modality employs specialized sensing electrodes to acquire these brainwave patterns from the scalp (Fig. 5 (b)). This acquisition process is the initial step, where multi-channel electrodes record raw analog EEG signals. This fundamental neuro-electronic interface captures the synchronized postsynaptic cortical activity, thereby laying the groundwork for interpreting the user's intentions [96], [97].

EEG signals are characterized by distinct frequency bands, known as EEG rhythms, each associated with different brain states and cognitive processes. As illustrated in Fig. 5 (c), these critical frequency bands typically include δ (0.5-4 Hz), θ (4-8 Hz), α (8-12 Hz), β (12-30 Hz), and γ (30-100 Hz) waves. For MI, specific rhythms such as α and β are particularly important for decoding the user's intentions as the power and characteristics within these bands are modulated by MI, providing the neural signatures that the processing pipeline aims to detect and interpret [37], [46].

The core of the MI-BCI involves a processing pipeline for

TABLE III
THE MOST COMMON TYPES OF BENCHMARK DATASETS USED FOR EVALUATING THE PERFORMANCE OF MI ALGORITHMS.

Cat.	Dataset	Channels	Sampling Rate	Subjects	Classes	Task
1	BCI Competition III dataset IIIa	25	250 Hz	9	4	Both hands, one foot, and tongue MI
2	BCI Competition III dataset IVa	118	100 Hz	5	3	Both hands and right foot MI
3	BCI Competition III dataset V	32	512 Hz	3	3	Both hands and word generation MI
4	BCI Competition IV dataset 2a	22 + 3 EOG	250 Hz	9	4	Both hands, both feet, and tongue MI
5	BCI Competition IV dataset 2b	3 EEG + 3 EOG	250 Hz	9	2	Both hands MI
6	Physionet EEG Motor Movement/Imagery	64	160/128 Hz	109	4	Both fists and both feet opening/closing MI
7	MI-OpenBCI	15	125 Hz	9	5	Both hands, both feet, and rest MI
8	Proprietary (Online)	–	–	–	–	–
9	High Gamma Dataset	44	250 Hz	14	5	Both hands, both feet, and rest MI
10	BCI Competition 2003 Dataset III	3	128 Hz	1	2	Both hands MI
11	BCI Competition II	2	128 Hz	3	2	Both hands MI

decoding the user's intentions that includes an analog front end (AFE), digital signal processing (DSP), and an interface to an actuator (Fig. 5 (d)). Starting from the AFE section, the AFE typically consists of mixed-signal circuitry that amplifies, filters (to remove noise and artifacts while preserving the relevant neural information), and digitizes EEG signals before digital processing [98], [99].

Following signal conditioning, the DSP stage prepares and interprets the EEG data to discern the user's intended motor command [37]. This complex process encompasses several critical steps: secondary signal preprocessing, which includes noise filtering and artifact removal to clean the data, followed by normalization to standardize signal amplitudes. Subsequently, segmentation divides the continuous data stream into meaningful epochs. The pipeline then moves to feature extraction and feature reduction, where techniques like power spectral analysis (e.g., calculating power in sensorimotor μ rhythm) are employed to derive salient characteristics from the EEG signals. Feature selection further refines these features, identifying the most discriminative ones. Finally, a feature classification algorithm takes these selected features and translates them into a specific motor command [35], [100]–[102].

The Actuator Interface serves as the crucial bridge between the processed neural data and the external device, translating the decoded motor intentions into physical actions. This stage begins with feature translation, where the output from the DSP stage (the classified motor command) is converted into a format understandable by the subsequent hardware [103], [104]. When analog actuation is required, a digital-to-analog converter (DAC) converts a digital command into an analog signal. To ensure safety and protect both the user and the equipment, an isolator galvanically separates the BCI system from the actuator, preventing any electrical interference or dangerous current flow. Following isolation, a filter may be applied to the analog signal to smooth it or remove any residual noise introduced during the conversion process. Finally, an amplifier boosts the signal to the required level to effectively drive the actuator, whether it's a motor, a prosthetic limb, or another output device [105].

Experimental paradigms for MI-BCI are designed to elicit distinct and detectable neural patterns associated with the imagination of movement. As illustrated in Fig. 5 (e), these paradigms commonly include tasks like imagining left/right hand movements, shoulder flexion/extension/abduction, and execution/imagination of fist opening/closing for both left and right hands. Other paradigms involve finer movements such as finger MI, finger tapping, left/right leg MI, and even cue-based tongue movements. The specific choice of paradigm influences the brain regions activated and the resulting EEG signatures [104], [106].

MI signal processing techniques can be evaluated using previously recorded EEG signals (offline) or by recording brain signals (online). To facilitate research and development, several publicly available datasets exist for MI experiments, as tabulated in Table III. In this table, the number of channels, sampling rate, number of subjects, number of classes, and the performed tasks are specified for each dataset. Despite these differences, most of these datasets are based on experimental paradigms involving the MI of the right hand, left hand, feet, and tongue [107]. These datasets constitute essential benchmarking resources for training and validating MI-BCI systems [108]. A prominent example of publicly available MI datasets is the BCI Competition, which has historically provided standardized data from various MI tasks, allowing researchers worldwide to benchmark algorithms and compare results under consistent conditions [109].

Based on Table III, the number of electrodes employed in MI-BCI systems ranges from 1 to 64 or more. In particular,

several studies reduce the number of electrodes originally provided by the acquisition devices [110]–[112]. While the traditional 'C3,4,z' set located over the central cortical area is often considered to reduce data dimensionality [55], [113], electrode selection is frequently constrained by the specific device used [114], [115]. Some works focus on specific central channels like 'C3,4' [113], 'Cz' [116], or 'C1,2' [58]. Researchers also frequently incorporate fronto-central and central-parietal channels, or extend the 'C3,4,z' group with additional electrodes such as 'FC3,4' [117], 'Fpz', and 'Pz' [118]. Other studies adopt specialized configurations, including 'Cz' with 'CP1,z,2' [119], or various subsets of 8, 9, or 10 electrodes to balance dimensionality reduction and cortical coverage [120], [121]. Furthermore, several datasets and authors explicitly report the sampling rates used during signal acquisition, which range from 125 Hz [56], 128 Hz [111], [122], 250 Hz [28], [123], 256 Hz [124], and 500 Hz [110], [116], [119], [125], [126] up to 512 Hz [127].

Finally, the decoded intention is translated into feedback mechanisms delivered to the user, such as vibration, tactile, thermal, electrical, haptic, auditory, or visual cues, completing the closed-loop system [110], [128], [129], as illustrated in Fig. 5 (f).

The DSP stage (shown in Fig. 5 (d)) is the computational core of the MI-BCI systems and has been the primary focus of research over the past decade. The following section provides a comprehensive review of the signal processing techniques employed at each stage of this pipeline.

V. MI-BCI Signal Processing Algorithms

Signal processing plays a central role in transforming raw EEG recordings into discriminative features that can be reliably used for MI-BCI applications. EEG signals are inherently complex, non-stationary, stochastic, and heavily contaminated by noise and artifacts, which makes the extraction of discriminative patterns a challenging task [130]. To overcome these limitations, multichannel EEG acquisition is often used to provide richer spatial information from the motor cortex, improving classification performance by offering a more detailed representation of neural dynamics [131]. However, this comes at the cost of increased data dimensionality and computational load, requiring sophisticated algorithms capable of efficiently handling high-volume, high-variance inputs. Because MI-BCI systems must operate with minimal latency, especially in real-time applications such as robotic control or neuroprosthetic movement, robust and optimized signal processing algorithms are essential to ensure both accuracy and responsiveness throughout the decoding pipeline [132], [133].

The processing of EEG MI signals typically involves several key stages, as illustrated in Fig. 5. First, pre-processing is crucial for mitigating noise and artifacts while enhancing EEG signal quality. Typical preprocessing operations include filtering (e.g., band-pass filtering to isolate relevant frequency bands like α and β) and artifact removal techniques [30], [134]–[137]. Following pre-processing, feature extraction aims to extract the most informative components from the cleaned EEG data. This stage can also involve techniques for feature reduction to decrease dimensionality and computational load, and feature selection to identify the most discriminative features that most effectively represent the user's MI intent [52], [138]–[142]. Finally, feature classification employs classical or AI-based algorithms to map the extracted features to specific motor commands, decoding the user's intended movement into actionable control signals.

A structured overview of impactful EEG-based MI-BCI papers over the past decade, as well as new papers with novel ideas, is tabulated in Table IV. For each paper, the publication year, pre-processing technique, feature extraction method (including any feature reduction or selection strategies), feature classification algorithm, dataset category (defined in Table III), and obtained performance metrics are reported. The commonly reported performance metrics in this table are the best or average obtained classification accuracy (CA), overall cross-subject classification accuracy (CSCA), overall cross-task classification accuracy (CTCA), intra-subject classification accuracy (INACA), inter-subject classification accuracy (INECA), subject-dependent classification accuracy (SDCA), subject-independent classification accuracy (SICA), F1-score, sensitivity (Sen), specificity (Spe), precision (Pre), Cohen's kappa coefficient (K), efficiency (Eff), recall (REC), standard deviation (STD), mean difference (MEAD), and p-value. In the next section, first, a brief review of AI-based signal processing is presented, and then the most common algorithms in Table IV are discussed.

A. Introduction to AI-Based EEG Signal Processing

Modern AI-based algorithms have dramatically enhanced MI signal processing by automatically extracting complex spatio-temporal features and improving CA and generalization. AI is the overarching field of creating intelligent systems; machine learning (ML), as illustrated in Fig 6, is a subset of AI where systems learn from data without explicit programming; neural networks (NNs) are a type of ML model inspired by the human brain; deep learning (DL) uses NNs with many layers (deep architectures) for more complex feature learning [98], [100], [183]; transformer models are a specific DL architecture excelling at sequential data processing, particularly useful for long-range dependencies in EEG [144], [184]; generative AI is a type of AI focused on creating new content; and large language models (LLMs) are a type of generative AI specialized in understanding and generating human-like text. All of these algorithmic families can be tailored directly or indirectly to learn discriminative patterns in MI EEG signals for robust feature extraction and accurate motor intention decoding [185], [186].

The ML algorithms are broadly classified into supervised, unsupervised, semi-supervised, and reinforcement learning, as depicted in Fig. 6. Supervised learning algorithms learn from labeled data to make predictions, widely used in MI signal processing for classifying MI tasks based on pre-labeled EEG signals. Unsupervised learning algorithms discover patterns in unlabeled data, applicable to MI for clustering different EEG states or identifying artifacts. Semi-supervised learning combines labeled and unlabeled data, offering a way to improve MI

TABLE IV
STATE-OF-THE-ART EEG-BASED MI-BCI SIGNAL PROCESSING ALGORITHMS REPORTED IN THE LITERATURE IN THE RECENT DECADE.

Ref.	Year	Pre-processing	Feature Extraction	Feature Classification	Dataset Category	Performance
[138]	2026	WPD, ICA, SWS	WVG	ConvNeXt	(4), (9)	Dataset (4): CA(90.20%), Spe(95%), Pre(87.8%), K(0.87). Dataset (9): CA(97.73%), Spe(98.3%), Pre(98.57%), K(0.94).
[143]	2026	BPF	MAML	MAML-LNN	(4),(6)	Dataset (4): 4-Class CA(81.3%), CTCA(76.3%), 4-Class K(0.75), CT K(0.69), 4-Class macro F1 (0.8), 4-Class macro F1 (0.8), CT macro F1(0.76). Dataset (6): 4-Class CA(79.6%), 4-Class K(0.729), 4-Class macro F1 (0.786). Eff: 0.82 M param., 0.17 G MACs, Speed: 38 fps.
[144]	2026	ICA, FIR BPF	TMEF-Net	TMEF-Net	(8)	INACA: 89%, INECA: 81%
[145]	2025	ZM-STD	MSCFormer	MSCFormer, FCL	(4),(5)	Dataset (4): CA(82.95%), K(0.7726). Dataset (5): CA(88%), K(0.7599).
[146]	2025	VAE	DAE	CNN-LSTM	(8)	CA: 94.36%, Pre: 0.94, Recall: 0.947, F1-Score: 0.94
[147]	2025	BPF, CSP, ICA	PSD (WCM), MRCP, ERD	SVM, LDA, KNN, RF	(8)	CA: SVM (80%), RF (79%), KNN (78%), LDA (60%) (k-fold cross-validation)
[134]	2024	ICA	TST	TST-FCL	(4),(5)	Dataset (4): SDCA TST-ICA(97.77%), Pre(1), REC(0.94), Spe(1), F1-score(0.97). Dataset (5): SDCA TST-ICA(87.29%), Pre(0.87), REC(0.84), Spe(0.88), F1-score(0.85).
[148]	2022	BPF, SWS	CNN LSTM	CNN LSTM FFN-FCL	(4)	CA: 94.1%, K: 0.921
[149]	2022	ZSN	CNN-TRF-POE	TRF-FCL	(6)	CA: 83.31% (2-class), 74.44% (3-class), 64.22% (4-class)
[150]	2022	PCM-GL	GCN	GCN-FCL	(6), (9)	Dataset (6): CA(93.06%), K(84.47%), Pre(88.39%), REC(88.35%), F1-score(88.34%). Dataset (9): CA(96.24%).
[53]	2022	CAR, ICA	FGANet	FGANet-WSPS	(8)	CA: 91.96%. STD: 5.82%
[141]	2022	FBTS, BPF	FBRTS	OVR-SVM	(4),(5)	Dataset (4): CA(77.7%), STD(13.6%), K(0.71). Dataset (5): CA(86.9%), STD(6.03%), K(0.7).
[151]	2022	BPF, ZSN	EEG Conformer	EEG Conformer	(4),(5)	Dataset (4): CA(78.66%), K(0.7150). Dataset (5): CA(84.63%), K(0.6926).
[140]	2021	BPF	CNN	CNN	(8)	SICA: 84.19%, STD: 9.98%
[152]	2021	OVR-FBCSP	OVR-FBCSP	HDNN-TL	(4)	HDNN-TL: K=0.81, HDNN: K=0.78.
[33]	2021	SWS, BPF	3D-CNN	3D-CNN	(4),(5),(7)	Dataset (4): CA(87.15%), STD(7.31%). Dataset (5): CA(75.85%), STD(12.80%). Dataset (7): CA(70.37%), STD(17.09%).
[153]	2021	BWF BPF	MIN2Net MTL	MIN2Net MTL	(4),(7)	Dataset (4): SDCA(65.23%), SD F1-score(64.72%), SICA(60.03%), SI F1-score(49.09%). Dataset (7): SDCA(61.03%), SD F1-score(63.59%), SICA(72.03%), SI F1-score(72.61%).
[82]	2021	MSPCA	EFD, IEFD	FFNN	(4),(5),(3),(11)	Dataset (4): SDCA(99.82%), Dataset (5): SDCA(93.33%), Dataset (3): SDCA(88.08%), Dataset (11): SDCA(91.96%), Dataset (4): SDCA(82.70%).
[154]	2021	NFL BPF	CSP, CSP, SW-LCR, SW-MODE	LDA	(4)	SW-LCR: CA(86.19%), STD(12.02%), K(0.72). SW-MODE: CA(86.03%), STD(12.04%), K(0.72).
[155]	2021	NFL BPF	TCSGL	TCSGL-FCL	(1), (4)	Dataset (1): CA(88.89%), K(0.7511). Dataset (4): CA(81.34%), K(0.8519).
[156]	2020	BPF	HS-CNN	HS-CNN	(4),(5)	Dataset (4): CA(91.57%). Dataset (5): CA(87.6%).
[157]	2020	BPF, WPT	CNN-CWT	CNN-CWT-TL	(10)	CA: 95.71%, K: 0.91
[158]	2020	BWF BPF , STFT	CNN-DCGAN	CNN-DCGAN	(4),(5)	Dataset (4): CA(83.2%), STD(3.5%), K(0.564). Dataset (5): CA(93.2%), STD(2.8%), K(0.677).
[159]	2020	BWF BPF	DRDA	DRDA	(4),(5)	Dataset (4): CA(74.75%), K(0.6634). Dataset (5): CA(83.98%),K(0.6796).
[160]	2020	BPF	PCA, FLDA	H-KELM	(1)	CA: 96.54%
[161]	2019	ESDA, FIR HMW BPF	CSP	LDA	(4),(5)	Dataset (4): CA(79.79%). Dataset (5): CA(67.75%).
[135]	2019	BPF, CSP, SSIG	CSP, MIN, CNN	CNN	(8)	SDCA: 71.32%. SICA: 74.15%.
[162]	2019	-	CP-MixedNet	CP-MixedNet	(4), (9)	Dataset (4): CA(74.6%). Dataset (9): CA(93.7%).
[163]	2019	NFL BPF	DWT-EMD-SampEn	SVM	(5)	CA: 95.1%
[164]	2019	STFT	STFT	VGG-16-TL	(5)	CA: 74.2%
[165]	2018	-	CNN	CNN-TL	(6)	CA: CNN(80.38%), CNN+TL(86.49%)
[166]	2018	BPF, SWS	CSP, TSGSP	SVM	(1),(4),(5)	Dataset (1): CA(88.5%), STD(12%). Dataset (4): CA(84%), STD(7.5%). Dataset (5): CA(84.3%), STD(15%).
[167]	2018	SWCS	FBCSP	RNN-GRU-LSTM	(4)	CA: 73.56%
[168]	2018	NFL BPF, CCA	DWT	TSK FNN	(6)	CA: 79.4%
[169]	2018	BPF	FBCSP	CNN	(4)	CA:74.46%
[170]	2018	BPF, LAP, CWT, CAR	2-Layer CNN	2-Layer CNN	(4)	CA: 85.59%, F1-score: 0.85, k: 0.766
[171]	2017	BWF BPF	AR-CSP	SRSG-FasArt	(4)	CA: 85.49%, STD: 11.64%, K:0.63
[172]	2017	BPF	FFT, 5-layer CNN	5-layer CNN	(8)	CA: 86.41%, STD: 0.77%
[173]	2017	BPF	FBCSP	CNN	(4)	CA: 84%
[174]	2017	BPF, MSPCA	WPD	KNN	(2)	CA: 92.8%
[175]	2017	BPF	CNN	CNN-SAE	(5)	K: 0.547
[176]	2017	BPF	ARM, WPD, SampEn, PCA	SVM	(8)	CA: 98%
[177]	2016	BWF BPF, SWS	FFT, WPD	FDBN	(5)	CA: 84%
[178]	2016	BPF	OAL	NVB	(5)	CA: 0.64%
[179]	2015	BPF	SFBCSP	SVM	(2)	CA: 91.05%, STD: 2.45%
[180]	2015	BWF BPF	ACSP	CNN	(4)	CA: 69.27%
[181]	2014	CAR, LAP, CSP	PCA	SVM	(4)	CA: 76.34%
[182]	2014	FFT, PCA	CDBN	CDBN	(1),(2),(10)	CA: 88.25%, STD:5.70%

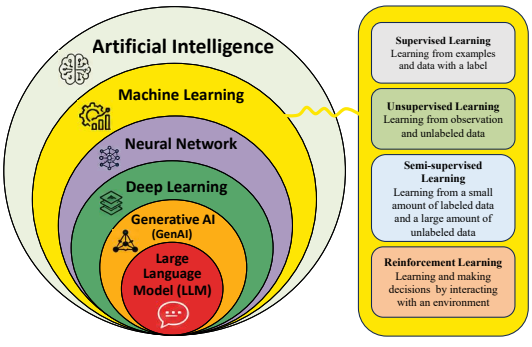

Fig. 6. The relation between various categories and sub-domains of artificial intelligence (AI).

models when labeled data is scarce. Reinforcement learning involves agents learning through trial and error by interacting with an environment, which can be used in MI-BCI systems for adaptive control and optimizing BCI performance over time [187]–[189].

Besides traditional signal processing algorithms, which are still being used in EEG because of their simplicity, AI-driven approaches have significantly advanced the performance and robustness of EEG-based BCI systems. Classic ML algorithms, such as support vector machines and linear Discriminant analysis, remain highly relevant and effective for MI signal processing [130], [147], [181], [190]. These algorithms excel at finding discriminative patterns within extracted EEG features, enabling accurate classification of MI intentions, particularly when combined with robust feature engineering techniques.

Specifically, as depicted in Fig. 7, DL-based architectures such as convolutional neural networks (CNNs) can simultaneously perform feature extraction and feature classification tasks, as opposed to traditional algorithms utilized for MI tasks [33], [102], [148]. Additionally, recurrent neural networks, transformers, and hybrid (HYB) models automatically learn hierarchical EEG features, reducing the need for manual pre-

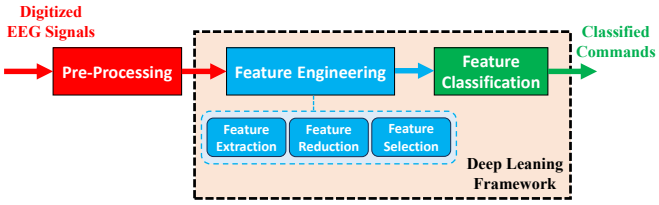

Fig. 7. Comparison of traditional processing of EEG-based MI signals and DL-based architectures.

processing and excelling at capturing complex temporal patterns, inter-channel relationships, and user-specific variations, leading to higher CA and better generalization [144], [149], [169]. Also, techniques such as positional embedding (POE) incorporate temporal position information into transformer inputs, enabling attention mechanisms to preserve the sequential order of EEG signals [149]. Moreover, while specialized models like EEGNet, ShallowConvNet, and DeepConvNet are tailored for efficiency and specific tasks like MI, newer models like EEG-TCNet, SincNet, EEG Conformer, CLTNet, and DB-BISAN integrate temporal convolutional networks, trainable sinc-filters, and self-attention mechanisms to handle temporal dynamics and long-range dependencies [184], [191]. Furthermore, the MSCFormer algorithm combines multi-scale CNN branches with a transformer encoder to jointly learn local spatial patterns and long-range temporal dependencies for end-to-end MI decoding [145]. These innovations collectively enable robust, low-latency, and power-efficient neural decoding necessary for real-time MI-BCI applications. In the following sections, a detailed discussion is provided on the most common algorithms, as listed in Table IV, for each signal processing block.

B. EEG Pre-Processing Algorithms

Signal pre-processing is a critical first step in the analysis of EEG signals, particularly for MI tasks. Its primary goal is to clean the raw EEG data by removing artifacts that can obscure the neural signals of interest. These artifacts can arise from various sources, including eye blinks, muscle movements (EMG), electrical noise from the environment, and impedance issues [98]. Effective pre-processing ensures that the subsequent feature extraction and classification steps operate on cleaner, more representative neural data, thereby improving the accuracy and robustness of motor intention decoding [137]. The most common EEG signal pre-processing algorithms for MI systems can be categorized into four general groups, as shown in Fig. 8.

1. Blind Source Separation Methods: The Blind source separation (BSS) methods operate by decomposing multichannel EEG into statistically independent or decorrelated source signals, allowing artifact and neural components to be isolated without prior knowledge of their origins. The independent component analysis (ICA) is the most widely used technique of this class and decomposes EEG into statistically independent components to identify and remove artifacts (e.g., eye blinks, muscle activity, cardiac noise) [134], [147]. Moreover, a deep convolutional generative adversarial network (DCGAN) learns latent source representations, and as a generative model, it can separate or reconstruct independent EEG components from mixed signals [72], [158]. Other techniques in this group are principal component analysis (PCA), canonical correlation analysis (CCA), and some other spatial filtering methods [190], [192]. The adaptive sparse representation (ASR) has also been employed for artifact suppression by representing EEG signals using sparse bases while preserving neural components [193].

2. Transform-based Methods: The transform-based methods analyze EEG signals in joint time-frequency domains. The wavelet transform (WT) is one of the most commonly used algorithms in this group, providing a flexible, multiresolution decomposition that captures transient MI-related oscillatory patterns more effectively than fixed-window transforms and is especially useful because MI-related rhythms (μ and β) are nonstationary [194], [195]. Also, wavelet packet transform (WPT) decomposes EEG signals into multiple frequency subbands, such as the μ and β rhythms, providing finer frequency resolution than conventional wavelet decomposition [157]. The spectral-spatial input generation (SSIG) method further enhances this process by transforming raw EEG into combined spectral and spatial representations, allowing classifiers to exploit both frequency-specific activity and the spatial distribution of cortical rhythms [135]. Furthermore, spatio-temporal formatting (STF) extends this idea by restructuring EEG into unified spatio-temporal feature maps, preserving channel relationships and temporal dynamics simultaneously, which improves compatibility with DL models that require structured, image-like inputs [172].The empirical Fourier decomposition (EFD) and its improved variant, IEFD, have recently been introduced to adaptively decompose nonstationary EEG signals into intrinsic Fourier components for improved time-frequency analysis [82]. The time-frequency distribution (TFD) techniques are also widely used to characterize the temporal evolution of MI-related spectral activity [196]. Finally, multilayer weighted visibility graph (WVG) converts EEG time series into weighted graph representations, preserving temporal relationships and structural characteristics that facilitate robust feature extraction from non-stationary signals [138].

3. Filtering Methods: This is a major class and includes several subclasses:

A) Frequency Filtering: Different types of filters can be included in this category, such as low-pass (LPF), high-pass (HPF), band-pass (BPF), and band-stop filters (BSF), such as FIR and IIR [197], [198]. Specifically, notch filtering (NFL) is a band-stop filter designed to suppress a specific narrow frequency band, typically 50 Hz or 60 Hz power-line noise and is widely used for EEG signals [30], [55], [95], [136], [190], [198]. Moreover, Butterworth filtering (BWF) is a smooth band-pass/band-stop filter commonly used for BPF for MI signals to isolate μ (8–13 Hz) and β (13–30 Hz) rhythms [14], [30], [147], [197], [199]–[204]. Sliding-window strategies, including SW-LCR and SW-Mode, are also employed to improve temporal localization and signal segmentation before feature extraction [154].

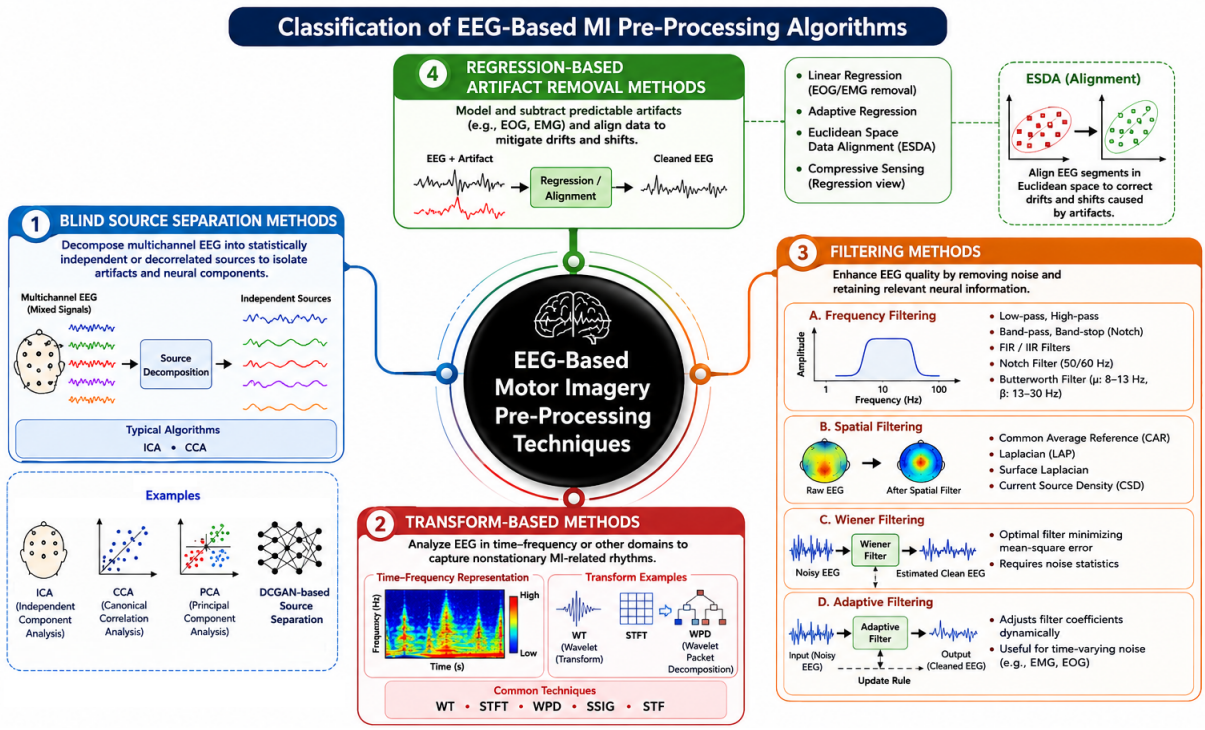

Fig. 8. Taxonomy of EEG pre-processing algorithms for MI-based BCI systems. EEG pre-processing techniques are broadly classified into blind source separation, transform-based, filtering, and regression-based artifact removal approaches. Typical algorithms and their corresponding signal processing mechanisms are illustrated, demonstrating how these methods enhance signal quality, reduce artifacts and noise, and facilitate reliable feature extraction and classification in MI-BCI applications.

The zero-mean standardization (ZM-STD) normalizes EEG signals by centering each channel around zero mean, reducing amplitude variability and improving the robustness of subsequent learning algorithms [145]. The Z-score normalization (ZSN) standardizes EEG features to zero mean and unit variance, reducing scale differences across channels and improving model convergence [149]. Additionally, sliding window segmentation (SWS) [138], [166], weighted overlap-add (WOLA), and sliding window cropping strategy (SWCS) techniques are also frequently employed to improve temporal and spectral resolution in EEG preprocessing [167]. Specifically, SWS divides continuous EEG signals into short, partially overlapping windows, ensuring local stationarity and reducing spectral smearing during subsequent analysis [154], [202]. The Hamming window (HMW) is also commonly applied during segmentation, smoothing window edges to reduce spectral leakage; for example, a 3.5 s EEG segment processed with a 50% overlapping Hamming window yields six 1 s frames for downstream analysis [205]. The WOLA algorithm builds on similar principles, applying weighted windowing and overlap reconstruction to achieve smooth transitions between segments while preserving amplitude continuity [206]. Also, the SWCS method, in turn, refines this process by cropping and aligning windowed segments to focus on the most relevant time intervals related to MI, which helps suppress noise and enhance rhythmic components in the μ and β bands [207], [208].

B) Spatial Filtering: The spatial filtering methods enhance EEG signal quality by combining information across neighboring electrodes to suppress global noise and highlight localized cortical activity relevant to MI. For example, the common average reference (CAR) is a widely used spatial filtering approach that re-references each electrode to the average of all electrodes, reducing global noise and improving signal localization [136]. Additionally, Laplacian (LAP) is typically utilized to enhance spatial resolution by emphasizing local potentials relative to their neighbors, effectively acting like a high-pass spatial filter [30], [57], [95], [136], [181], [192], [199]. The artifact-rejected common spatial pattern (AR-CSP) extends conventional CSP by incorporating artifact rejection during spatial filtering, leading to more robust MI feature extraction [171].

C) Wiener Filtering: This technique offers an optimal approach to signal processing by minimizing the mean-square error between the estimated signal and the desired output under known noise characteristics, making it particularly effective for enhancing the signal-to-noise ratio in EEG recordings during MI tasks, where neural signals are often obscured by significant environmental and physiological artifacts [209].

D) Adaptive Filtering: This group of filtering techniques offers a dynamic advantage by adjusting filter parameters in real-time, making them highly useful for removing time-varying noise that is common in EEG signals during MI tasks, thereby improving the quality and reliability of the brain-computer interface [210].

4. Regression-based Artifact Removal Methods: This class includes techniques that model and subtract predictable artifacts (e.g., EOG, EMG) using linear regression or adap-

tive regression approaches [55], [90]. Among these, the Euclidean space data alignment (ESDA) methods are particularly more common by aligning EEG data segments in Euclidean space, essentially correcting for drifts and shifts that are often caused by artifacts [161]. This is formalized through the Euclidean space (EUS) representation, in which EEG covariance matrices or trial segments are aligned within a common Euclidean reference frame to reduce inter-subject and inter-session variability prior to alignment-based correction [57]. This alignment implicitly models and subtracts these predictable artifactual components, helping to preserve the underlying neural activity. Furthermore, while compressive sensing (CPS) is more commonly associated with blind source separation due to its sparsity-based unmixing, it can be viewed through a regression lens when considering the reconstruction of artifact-free signals from undersampled data, effectively regressing out the artifactual components [211].

C. EEG Feature Engineering Algorithms

In the Feature Engineering section of an EEG-based MI processing pipeline, as depicted in Fig. 7, feature extraction is the initial step where raw EEG data is transformed into a set of relevant features that capture the underlying characteristics of the MI. Techniques here aim to derive meaningful information from the signals [140], [180], [212]. Extraction may produce a very large set of features. Feature reduction techniques are then applied to decrease the dimensionality of this feature set while retaining as much of the important information as possible. This helps in combating the curse of dimensionality and improving the efficiency of subsequent steps. Finally, feature selection techniques are used to choose a subset of the most relevant features from the reduced set (or directly from the extracted features if reduction wasn't performed or was minimal). This step focuses on identifying the features that are most discriminative for the MI task, further improving model performance and reducing overfitting [102], [213].

The most widespread feature extraction algorithms can be broadly categorized into five groups based on the mathematical transformation applied to the raw EEG data to extract patterns, as illustrated in Fig. 9. EEG feature-extraction algorithms can be classified based on the mathematical domain of the signal representation into five foundational domains and a combined approach. The time-domain techniques analyze raw data sample points over time (P_A, P_2, Q_1) [176]. The frequency-domain techniques perform spectral analysis (PSD) to isolate distinct brain rhythms (δ, θ, α, β) [52], [200], [204]. The time-frequency domain techniques capture joint temporal and spectral dynamics [158]. The spatial domain techniques map source separation and localized cortical activity (e.g., C3, C4 topomaps) [154]. The feature selection/reduction techniques optimize the feature vector via dimensionality reduction [142]. Moreover, the HYB techniques leverage multi-domain integration and data-driven DL architectures (e.g., Wavelet + CNN) for enhanced classification.

1) Time-Domain Techniques: Time-domain techniques extract features directly from the raw EEG signal's voltage fluctuations over time, capturing characteristics like amplitude, variance, and shape, and encompass a range of techniques that analyze the signal's amplitude and statistical properties directly over time.

Simpler methods like root-mean-square (RMS) and integrated EEG (IEEG) quantify signal power and cumulative amplitude, respectively. More complex approaches involve modeling the signal's temporal dependencies: autoregressive models (ARM) capture this by predicting future signal values based on past observations, while adaptive autoregressive (AAR) models adjust these predictions over time [176], [196]. Adaptive filtering algorithms such as recursive least-squares (RLS), least mean squares (LMS), and the Kalman filter (KF) are also employed to estimate or track signal characteristics in the time domain, often for noise reduction or state estimation. The quaternion-based signal analysis (QBSA) method offers an advanced mathematical framework for analyzing multi-dimensional EEG data within the time domain, leveraging quaternion algebra to capture complex signal dynamics [51], [194].

In addition, statistical features such as standard deviation, skewness, kurtosis, entropy, and energy provide simple yet effective time-domain descriptors of EEG signals. These measures quantify the variability, asymmetry, peakedness, complexity, and power of the waveform, allowing the extraction of discriminative patterns related to MI activity. Furthermore, ERD/ERS-based algorithms are crucial time-domain techniques. ERD/ERS analysis quantifies changes in the power of specific frequency bands, typically the alpha and beta bands, in the EEG signal that are time-locked to a cognitive or motor event. ERD refers to a decrease in the power of brain oscillations, often observed during motor imagery execution, while ERS indicates an increase in power, which might occur after the task or in specific conditions. By analyzing these power modulations over time, ERD/ERS provides a powerful way to capture the neural dynamics underlying motor imagery [147], [214], [215].

2) Frequency-Domain Techniques: Frequency-domain techniques extract features from EEG MI signals by analyzing how signal power is distributed across different frequency bands, rather than in the raw time waveform. Methods such as the fast Fourier transform (FFT) and Welch's method (WCM) for power spectral estimation highlight characteristic oscillatory patterns, particularly in the μ and β rhythms, that change during imagined movements, making them effective for identifying MI-related neural activity [55], [177], [196]. Specifically, FFT efficiently converts a time-domain EEG signal into its constituent frequency components, as illustrated in Fig. 10. By analyzing the power spectral density across various frequencies, the FFT reveals the strength of oscillations in bands like alpha, beta, and gamma, which are critical for distinguishing between different MI states as these patterns shift with imagined movements [57]. Specifically, Welch's method power spectral density (WCM PSD) estimation averages periodograms over overlapping windowed segments, reducing variance in the spectral estimate and yielding more stable power-band features for MI discrimination [95]. Also, local characteristic-scale decomposition (LCD) decomposes EEG signals into adaptive intrinsic scale components, enabling

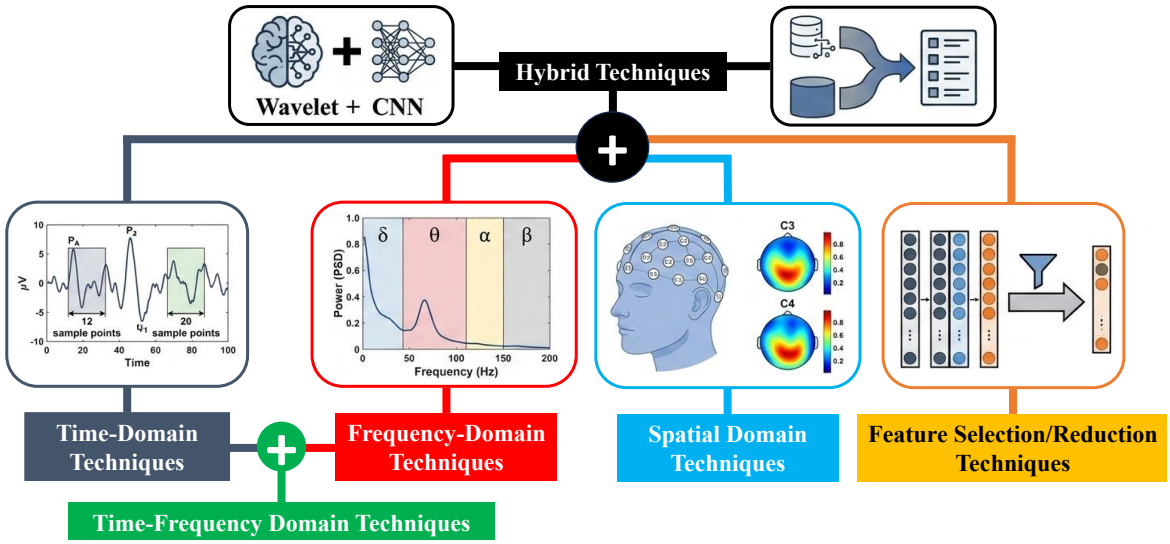

Fig. 9. EEG signals Feature Extraction algorithms classifications based on the mathematical domain of the signal representation. The classification spans five foundational domains and a combined approach: Time-Domain Techniques, Frequency-Domain Techniques, Time-Frequency Domain Techniques, Spatial Domain Techniques, Feature Selection/Reduction Techniques, and HYB Techniques.

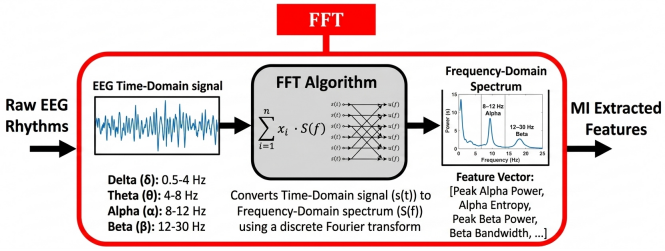

Fig. 10. Applying FFT conversion to EEG MI signals for extracting the delicate frequency features of MI oscillations. The FFT decomposes the raw, time-domain EEG rhythms into a frequency-domain spectrum, isolating power changes in target bands (e.g., alpha and beta rhythms) to construct a discriminative feature vector for classification.

the extraction of frequency-specific features useful for distinguishing MI patterns [204]. Additionally, mutual information (MIN) measures the statistical dependency between frequency-band features and class labels, helping select the most informative spectral components for MI discrimination [136], [192], [216]. The partial directed coherence (PDC) method further quantifies directional connectivity by estimating the direction and strength of frequency-domain interactions between EEG channels, offering insight into information flow across cortical regions during MI [199].

3) Time-Frequency Domain Techniques: Time–frequency domain techniques extract EEG features by capturing how signal frequency content evolves over time, making them well-suited for the non-stationary nature of MI signals. Methods like the short-time Fourier transform (STFT) provide a fixed time–frequency resolution by sliding a window across the signal, while wavelet-based approaches offer more flexibility through adaptive windowing [158], [164], [217].

The WT-based methods, especially the discrete wavelet transform (DWT) and continuous wavelet transform (CWT), are widely used in MI analysis because they decompose EEG into multiscale representations that localize transient oscillatory patterns linked to MI-related μ and β rhythms [155], [157], [196]. The DWT pipeline passes the non-stationary EEG rhythms through successive low-pass and high-pass filter banks to yield approximation and detail coefficients, capturing macro-level trends and rapid transients across multi-level sub-bands before compressing them into a concise feature vector [163], [168], [212], [218]. Conversely, the CWT continuously shifts and scales a selected mother wavelet across the signal to map out a high-resolution time-frequency scalogram [170]. Specifically, in DWT, the raw EEG rhythm is passed through a multi-level filter bank to decompose the signal into low-frequency approximation (A_1) and high-frequency detail (D_1, D_2) coefficients, which are reduced to form a structured feature vector, as illustrated in Fig. 11 (a). In CWT, the signal is convolved with continuous mother wavelets (e.g., Morlet, Meyer) to generate a high-resolution time-frequency scalogram, from which energy variations within critical motor imagery bands (alpha and beta) are selected as features, as depicted in Fig. 11 (b).

Wavelet packet decomposition (WPD) extends this capability by further splitting both approximation and detail coefficients, allowing finer and more comprehensive analysis of time-frequency components [174], [176], [177], [201]. Empirical mode decomposition (EMD) offers a data-driven alternative by adaptively decomposing the EEG signal into intrinsic mode functions without requiring a predefined basis, making it well suited to the nonstationary, nonlinear nature of MI rhythms [192]. Additionally, spatio-spectral feature representation (SSFR) integrates spatial filters with time-frequency features to enhance the extraction of MI-related patterns [33], while the separable common spatio spectral pattern (SCSSP) algorithm jointly optimizes spatial and spectral filters in a separable manner to improve class discrimination [136], [219].

4) Spatial Domain Techniques: Spatial-domain feature extraction methods focus on enhancing the discriminative

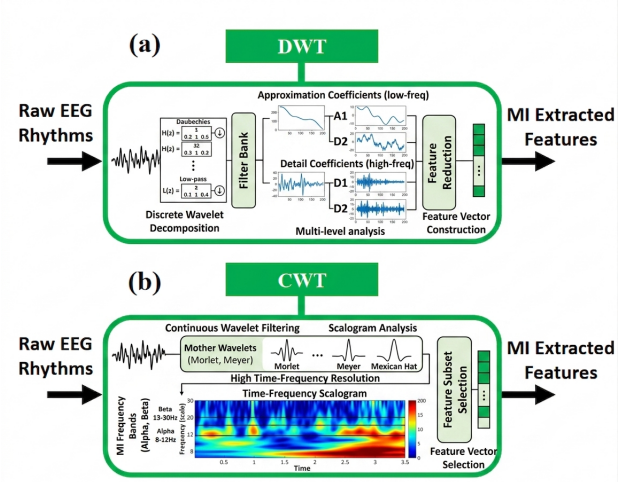

Fig. 11. EEG wavelet transform-based feature extraction methods. Applying DWT (a) and CWT (b) to MI signals.

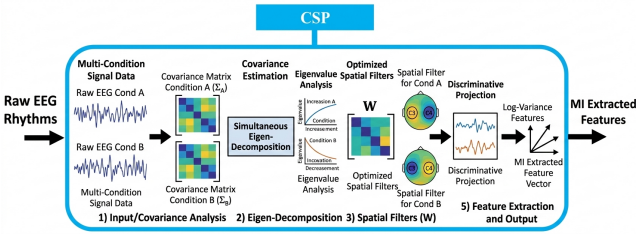

Fig. 12. Utilizing CSP-based methods for EEG MI signal feature extraction.

spatial patterns of EEG signals by exploiting the distribution of neural activity across different electrodes, with common spatial patterns (CSP) being the most widely used approach for MI [130], [154], [181], [204], [220], [221]. CSP learns spatial filters that maximize variance differences between MI classes, effectively highlighting the sensorimotor activation changes characteristic of left-or right-hand imagery [142], [159], [182], [222]. Utilizing CSP-based methods for EEG MI signals feature extraction can be described in a few steps, as illustrated in Fig. 12. (1) Input and Covariance Analysis: Multi-condition EEG signal data (Condition A vs. Condition B) are used to estimate the respective spatial covariance matrices ($\sum A$ and $\sum B$). (2) Simultaneous Eigen-Decomposition: Joint diagonalization is performed via eigenvalue analysis to maximize the variance of one condition while minimizing the variance of the other. (3) Spatial Filters (W): An optimized spatial filter matrix (W) is constructed, corresponding to specific target cortical regions (e.g., C3 and C4 topomaps). (4) Discriminative Projection & Feature Extraction: The raw EEG signals are projected onto the subband spatial filters to yield discriminative time-series waveforms, from which the log-variance is calculated to form the final optimized MI-extracted feature vector. To better preserve electrode topology in spatial-domain analysis, the azimuthal equidistant projection (AEP) method approximates the scalp as a sphere and maps electrodes from three-dimensional space onto a two-dimensional plane, facilitating image-like spatial representations for downstream CNN processing [205]. Furthermore, the filter bank Riemannian tangent space (FBRTS) combines filter-bank processing with Riemannian geometry to extract discriminative covariance-based representations from multiband EEG signals [141].

Its extensions, such as the common spatio-spectral pattern (CSSP) and common sparse spatio-spectral pattern (CSSSP), further integrate spectral information and sparsity constraints to improve robustness [166], [216]. The sub-band CSP (SBCSP) enhances performance by applying CSP to frequency-filtered sub-bands, capturing finer MI-related rhythmic variations, while methods like the temporally constrained sparse group spatial pattern (TSGSP) enforce structured sparsity to reduce noise and improve generalization [166], [223]. Although simpler, PSD can also serve as a spatial-domain descriptor when computed across electrodes, complementing CSP-based techniques in MI classification [95], [200]. Additionally, the augmented CSP (ACSP) technique extends traditional CSP by incorporating auxiliary or reference data to stabilize spatial filter learning, thereby improving robustness under non-stationary MI conditions [180]. Other representative feature extraction techniques include permutation entropy (PME) [224], sample entropy (SampEn) [224], movement-related cortical potentials (MRCP) [147], utilizing sensorimotor rhythms (SMR) [37], Pearson correlation matrix-based graph Laplacian (PCM-based GL) [150], signal-space modeling (SSM) [37], and eigenvector methods (EVM) [196].

5) Feature Selection/Reduction Techniques: Feature selection and reduction techniques aim to eliminate irrelevant or redundant information while preserving the most discriminative patterns for MI classification [142]. Dimensionality-reduction algorithms such as PCA and ICA transform high-dimensional EEG features into compact representations by maximizing variance or statistical independence, respectively, thereby improving computational efficiency and robustness [82], [160], [181], [217]. The Canonical variate analysis (CVA) selects user-specific features that best reflect the MI task for each individual, which can then be used to train subject-tailored classifiers [95]. Filter bank approaches like filter bank CSP (FBCSP) [131], [173], [192], [225], discriminant filter bank CSP (DFBCSP), and sparse filter bank CSP (SFBCSP) further refine feature spaces by decomposing EEG signals into multiple frequency sub-bands and applying spatial filters optimized for discrimination [167], [198], [216], [222]; these variants enhance class separability through discriminant weighting (DFBCSP) or sparse regularization (SFBCSP), enabling more informative and noise-resistant feature sets [141], [179]. This is enabled through filter bank decomposition (FBD), in which a bank of bandpass filters splits the EEG signal into multiple sub-bands that serve as parallel inputs to subsequent spatial filtering stages such as CSP [223]. The one-versus-rest FBCSP (OVR-FBCSP) extends FBCSP to multi-class MI problems by learning class-specific spatial filters for each one-versus-rest classification task [152]. Moreover, an iterative algorithm (ITA) approach selects frequency bands by iteratively evaluating candidate bands and retaining the one that yields the highest CA for a given binary MI task [198]. Furthermore, the multiscale PCA (MSPCA) extends PCA across multiple scales to simultaneously perform denoising and dimensionality

reduction [82].

A second category involves evolutionary and population-based optimization algorithms, which search for the most effective subset of features or model parameters through iterative, bio-inspired processes [221]. Methods such as particle swarm optimization (PSO), differential evolution (DE), artificial bee colony (ABC), ant colony optimization (ACO), genetic algorithm (GTA), and firefly algorithm (FFA) evaluate candidate feature subsets based on classification performance and evolve them toward optimal solutions using mechanisms like swarm behavior, mutation, selection, pheromone trails, or light-intensity attraction [204], [221]. Techniques like ACO and simulated annealing (SAN) are particularly useful for navigating large combinatorial feature spaces; ACO constructs probabilistic solution paths guided by learned pheromone levels, while SAN explores feature subsets using a temperature-controlled search strategy that helps escape local optima [221]. The learning automata (LAM) method, often combined with the firefly algorithm, have also been proposed to optimize feature selection for MI EEG by adaptively updating action probabilities based on environmental feedback [204]. The HYB approaches combine these evolutionary strategies with dimensionality-reduction or domain-specific feature extraction methods, achieving more efficient searches and often producing superior, task-adaptive feature subsets for MI-BCI systems [223], [226].

Beyond population-based approaches, sparsity-driven and task-aware learning techniques provide additional mechanisms for targeted feature reduction. The joint sparse optimization (JSO) enforces shared sparsity across multiple feature vectors or sub-bands, promoting the selection of consistent, physiologically meaningful features while simultaneously suppressing noise [166], [227]. The optimal allocation (OAL) selects representative EEG trials with minimal variability, improving the quality of training data and enhancing classification robustness across subjects. Furthermore, the multitask learning (MTL) extends this idea by jointly learning feature representations across related MI tasks, sessions, or subjects, enabling knowledge transfer and improving robustness in the presence of inter-subject or temporal variability. These methods not only enhance generalization but also help identify stable neural signatures of motor imagery, making them valuable components in advanced MI-BCI pipelines [29].

6) Hybrid, Data Driven, Deep Learning Approaches: These feature extraction methods integrate information from multiple domains to capture the complex, non-stationary structure of EEG signals more effectively than traditional single-domain approaches. In practice, many studies combine handcrafted features from several domains, such as time-frequency features, spatial filters, or spectral measures, with learned representations from data-driven models to create richer and more robust feature vectors, ultimately improving CA in MI tasks [159]. Recent studies have proposed specialized deep architectures, including graph convolutional neural networks (GCNs) [150], deep autoencoders (DAEs) [146], multitask learning autoencoders (MTL-AECs) [153], transformer-based multi-branch EEG fusion networks (TMEF-Net) [144], and channel-projection MixedNet (CP-MixedNet) [162] for learning more discriminative spatial-temporal EEG representations. Specifically, DAEs and MTL-AECs have also been employed to learn discriminative latent representations (DLRs) that improve feature compactness and class separability for MI decoding [153]. The HYB EEG-fNIRS frameworks have also been introduced by converting multimodal signals into 3D tensors (EEG/fNIRS-TC) and employing fNIRS-guided attention layers (FGAL) to enhance cross-modal feature fusion [53]. Also, FGANet performs multimodal EEG-fNIRS fusion by using fNIRS-guided attention to emphasize informative brain regions while combining complementary neural and hemodynamic features [53]. The MIN2Net is a multi-task autoencoder that jointly performs latent feature learning, deep metric learning, and classification to obtain compact and discriminative EEG representations [153]. Furthermore, deep representation-based domain adaptation (DRDA) performs domain adaptation by learning deep representations that minimize distribution discrepancies between source and target domains, improving cross-subject generalization [159].

The HYB combinations like DWT-AR, which apply DWT for multiresolution decomposition followed by AR modeling to characterize temporal dynamics within each sub-band, provide a compact yet informative representation of MI-related rhythms. Similarly, Hilbert transform (HLT)-DWT methods use the HT to extract instantaneous amplitude or phase information from DWT-derived components, enhancing the sensitivity to transient oscillatory changes critical for MI classification [228]. Moreover, CNNs play a key role in this category because they can implicitly learn spatial, temporal, and frequency patterns simultaneously from raw or minimally processed EEG, even though they do not fit neatly into classical time, frequency, or spatial domain definitions [145], [156], [162]. Since CNNs also serve as classifiers, their detailed discussion is presented in the next section.

D. EEG Feature Classification Algorithms

EEG-based MI classification algorithms form the core of BCI systems designed to decode users' imagined movements from neural activity. These algorithms process non-invasive EEG signals to identify distinct patterns associated with different MI tasks, such as left- or right-hand movement imagination. The primary goal is to translate complex, noisy, and highly individual EEG data into accurate, real-time decisions, enabling effective interaction between the user and external devices such as prosthetics or neuro-rehabilitation tools. In the following, the most common classification algorithms based on Table IV are discussed in several categories, as depicted in Fig. 13. Note that Some algorithms naturally fall into more than one group.

1) Supervised Learning

The supervised learning algorithms extract spatial and temporal features from EEG signals to accurately discriminate motor intentions across subjects and sessions, where labeled EEG data, each corresponding to specific imagined movements, are used to train predictive models. These algorithms learn mappings between neural signal patterns and movement classes to

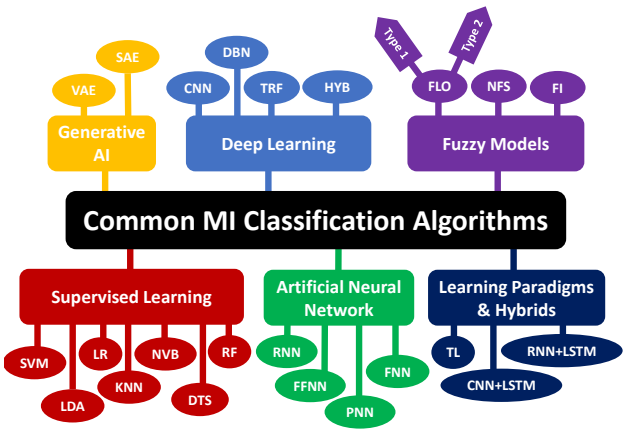

Fig. 13. Summary of prevalent AI-based MI feature classification algorithms.

enable real-time BCI control [188], [189]. The most common supervised approaches utilized for MI classification can be categorized as follows:

A) Linear & Margin-Based Classifiers: The linear and margin-based classifiers learn decision boundaries that separate classes using linear projections or maximum-margin principles, making them efficient and well-suited for high-dimensional EEG feature spaces. Their simplicity, interpretability, and compatibility with feature-selection or regularization strategies make them common baselines and reliable components in many MI-BCI pipelines [123], [189].

The support vector machines (SVMs) and their variants are among the most common and effective classifiers in this group used in MI-BCIs because they handle high-dimensional EEG features and small training sets exceptionally well [136], [147], [163], [179]. Standard SVM finds a maximum-margin separating hyperplane, making it robust to noise and inter-trial variability in MI signals [142], [166]. SVM classifies MI EEG signals by finding the optimal separating hyperplane that maximizes the margin between different MI classes. After extracting features, SVM maps them into a space, linear or kernel-transformed, and determines the boundary that best separates imagined movements, allowing robust and accurate discrimination even in noisy, high-dimensional EEG data [176]. Moreover, hybrid-kernel extreme learning machine (H-KELM) combines multiple kernel functions to improve nonlinear decision boundaries while maintaining fast training [160].

The complete pipeline of an EEG-based MI-BCI system utilizing the SVM for classification is illustrated in Fig. 14. The process begins with data acquisition, where raw EEG signals are recorded from the motor cortex during the imagination of distinct tasks, such as right-hand versus left-foot movements. These non-stationary signals are then converted into discriminative features, typically through frequency-domain methods like PSD, mapping the data into a low-dimensional feature space. Finally, a linear SVM algorithm is applied to perform binary classification. The model iteratively trains to find the optimal separating hyperplane that maximizes the geometric margin between the two classes. This decision boundary is uniquely determined by the support vectors, which are the critical data points closest to the hyperplane, ensuring robust generalization for real-time BCI applications.

For multi-class decoding, the one-versus-rest SVM (OVR-SVM) strategy extends this framework by training one classifier per class, offering a simple and computationally efficient solution [141]. The twin SVM (TSVM) further improves scalability by learning two nonparallel hyperplanes, significantly reducing training time while maintaining strong discriminative power. Optimized TSVM variants, such as GWO-TSVM, automatically tune critical parameters using metaheuristic search, enhancing accuracy and stability across subjects. Therefore, these SVM-based approaches form a highly reliable and widely adopted family of classifiers for MI features [229].

The linear discriminant analysis (LDA) family encompasses a group of linear classifiers designed to project high-dimensional EEG features onto a subspace that maximizes class separability [130], [136], [147], [199], [201], [216]. The LDA itself operates by finding a linear combination of features that best separates two or more MI classes (e.g., left vs. right hand). It assumes each class follows a Gaussian distribution with equal covariance, deriving a decision boundary that minimizes within-class variance while maximizing between-class variance [37], [154], [206], [217], [221].

Utilizing LDA-based methods for MI feature classification can be described in few steps, as shown in Fig. 15. (1) Initial Feature Space: Input MI extracted features initially exhibit high within-class variance and low between-class variance, making direct separation difficult. (2) Linear Projection: The algorithm computes an optimal discriminant vector (w) that maximizes the ratio of between-class variance to within-class variance, projecting the multi-dimensional features onto an optimized 1D axis. (3) Classification & Decision: The reduced-dimensionality space yields maximum class separation, allowing a precise classification threshold to split the distributions and output final discrete MI commands.

The fisher discriminant analysis (FDA) is conceptually similar and derived from the same statistical principle: it seeks a projection vector that maximizes the Fisher criterion, the ratio of inter-class scatter to intra-class scatter, yielding an optimal linear discriminant function. The linear discriminant classifier (LDC) employs the same underlying statistical foundations but emphasizes its implementation as a direct classifier, often using estimated covariance matrices to classify EEG feature vectors based on posterior probability [225]. Fisher's linear discriminant analysis (FLDA) is frequently paired with FBCSP-derived features, where the discriminant classifier's output accuracy directly reflects the BCI system's overall MI-detection performance [192]. The Spectral regression discriminant analysis (SRDA) method reformulates LDA as a regression problem, substantially reducing computational cost for high-dimensional EEG feature spaces while retaining discriminative power [204]. Finally, LDA–SVM represents a HYB model in which LDA first reduces the feature dimensionality and enhances linear separability, and the SVM performs the final classification, typically leading to improved robustness and generalization in MI tasks while retaining LDA's linear interpretability [216], [221].

The logistic regression (LR) is a statistical model used for

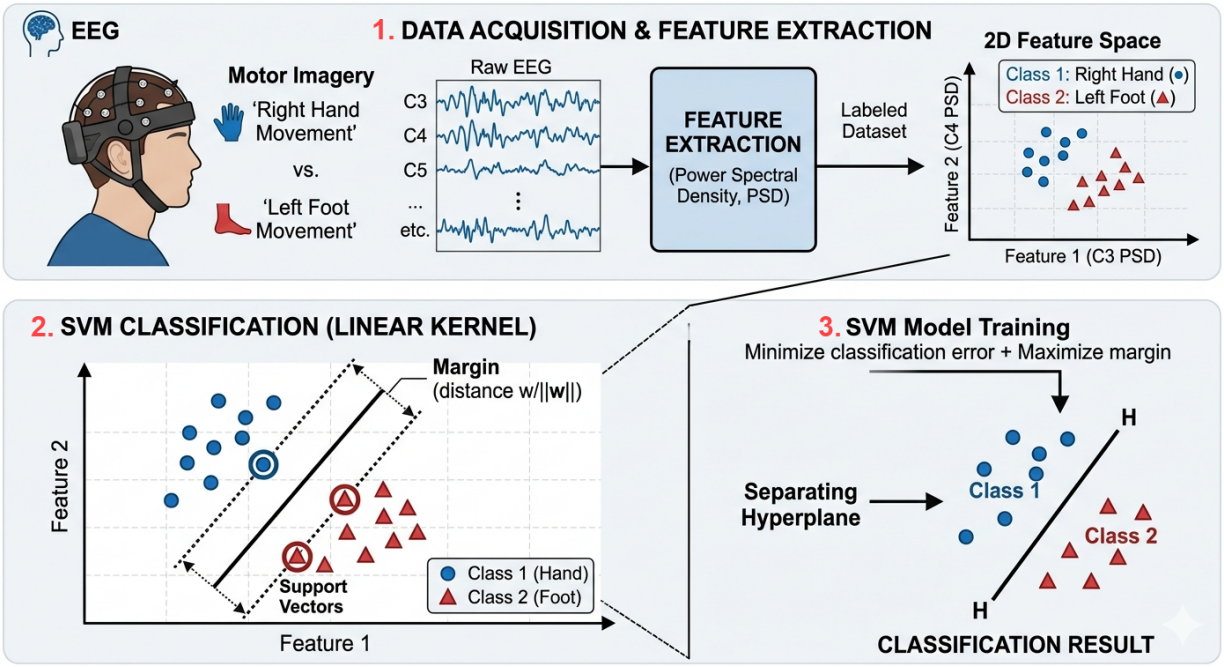

Fig. 14. Schematic workflow of an EEG-based MI-BCI classification system using the SVM algorithm. The Data Acquisition & Feature Extraction shows the raw EEG signals are recorded from motor cortex channels (e.g., C3, C4) during imagined right-hand vs. left-foot movements, and transformed into a 2D feature space using the PSD. The SVM Classification shows that a linear kernel SVM, for example, identifies the optimal separating hyperplane (H) by maximizing the margin between the two MI classes, uniquely defined by the critical edge data points known as support vectors.

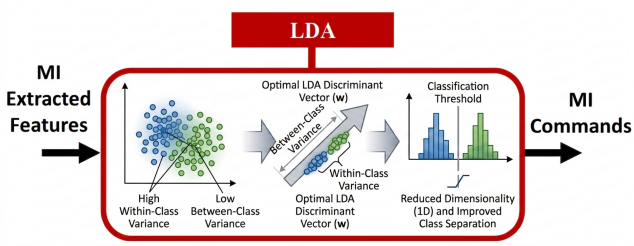

Fig. 15. Utilizing LDA-based methods for MI feature classification.

MI binary classification. It models the probability of a specific outcome (e.g., a particular MI class) using a logistic function applied to a linear combination of input features. This results in a decision boundary that is linear in the feature space, making it a simple yet effective linear classifier for differentiating between two MI states [230]. Moreover, the weighted logistic regression (WLR) introduces class-dependent weighting to improve robustness under imbalanced EEG datasets [220].

B) Instance- and Distance-Based Classifiers: The instance- and distance-based classifiers classify new MI samples by comparing them directly to stored training instances. Classification is determined by the proximity of a new sample to existing examples, typically using distance metrics like Euclidean or Mahalanobis distance. These methods rely on the idea that similar MI patterns occupy nearby regions in the feature space [231].

The k-nearest neighbor (KNN) is a non-parametric, instance-based learning algorithm. It classifies an unknown EEG feature vector by comparing it to the closest training samples in the feature space, typically measured using Euclidean or another distance metric [174], [232]. The algorithm then assigns the class most common among these neighbors to the test sample. In MI tasks, KNN leverages spatial or spectral EEG features, relying on the assumption that samples from similar MI patterns lie close together in the feature space [147], [221]. Additionally, the Riemannian classifiers use distance geometry principles to classify MI EEG signals by operating directly on covariance matrices viewed as points on a Riemannian manifold. Instead of relying on Euclidean distance, they measure similarity using Riemannian metrics that capture the intrinsic geometry of the data, enabling robust discrimination of MI patterns through spatial covariance structures [141], [221].

Utilizing KNN-based methods for MI feature classification can be performed in a few steps, as depicted in Fig. 16. (1) Feature Space Mapping: Input MI extracted features are mapped onto a multi-dimensional feature space containing labeled reference profiles (Condition A vs. Condition B). (2) Distance Measurement: When an unknown test point (star) is introduced, the algorithm calculates the spatial distance $d(\text{test}, \text{point}_i)$ to locate its nearest neighbors ($K = n$). (3) Majority Voting & Command Assignment: A local neighborhood decision is made in which the test point is assigned to the predominant class via majority voting (e.g., 2 votes for Condition A vs. 1 vote for Condition B), translating the feature data into a final discrete MI command.

C) Tree Based Methods: The tree-based methods capture nonlinear relationships in MI features and classify them by recursively partitioning the feature space into decision regions based on feature thresholds. Each split aims to maximize class

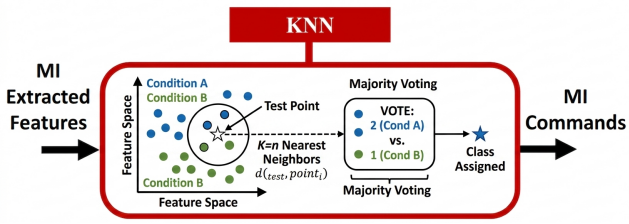

Fig. 16. Utilizing KNN-based methods for MI feature classification.

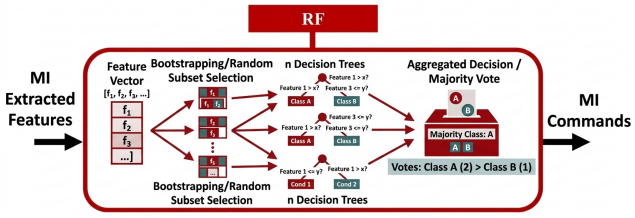

Fig. 17. Utilizing RF-based methods for MI feature classification.

purity, forming a hierarchical tree that predicts EEG class labels at its leaves.

Specifically, random forest (RF) is an ensemble learning method that classifies MI features by combining the outputs of many decision trees trained on different bootstrapped subsets of the data and features [233]. Each tree independently learns nonlinear decision rules that map MI-related features, such as band-power or covariance patterns, to class labels [147]. During prediction, the forest aggregates the votes of all trees (majority voting), which reduces overfitting and improves robustness to noise, making RF well-suited for handling the variability inherent in MI EEG recordings [234]. Utilizing RF-based methods for MI feature classification includes the following steps, as illustrated in Fig. 17. (1) Bootstrapping and Random Subset Selection: The input multi-dimensional MI feature vector $[f_1, f_2, f_3, \dots]$ undergoes bootstrap sampling to create distinct subset combinations, ensuring diversity across the ensemble. (2) Ensemble of Decision Trees (n Trees): Multiple independent decision trees are trained in parallel, where internal nodes evaluate specific threshold conditions (e.g., $Feature\,1 > x?$) to sequentially route data to candidate motor imagery classes (e.g., Class A or Class B). (3) Aggregated Decision and Majority Voting: The individual predictions from all n trees are collected and aggregated via a majority voting scheme (e.g., Votes: Class A (2) ¿ Class B (1)), combining weak estimators into a robust, unified output to generate the final target MI command.

The decision tree structure (DTS) classifies MI data by hierarchically splitting the feature space into regions using threshold rules that maximize class purity at each node, forming an interpretable tree that maps EEG features to MI classes. Also, the regression tree (RGT) follows a similar structure but predicts continuous outputs, such as class probabilities or feature scores, by fitting simple regression models within each leaf, enabling finer modeling of MI signal relationships compared to purely categorical splits [233]. Furthermore, the gradient boosting models (GBMs) and RGTs have also been investigated as ensemble learning approaches for MI classification [221], [235].

D) Probabilistic Models: The probabilistic models classify MI features by estimating the likelihood of each class given the observed features, using statistical distributions (e.g., Gaussian models or Bayesian inference) to capture uncertainty and variability in brain patterns for more flexible decision making [225]. For example, naive bayes (NVB) is a probabilistic classifier that models MI EEG features under the assumption of conditional independence, computing class-posterior probabilities from simple likelihood estimates, which makes it fast, lightweight, and effective even with limited training data. A Bayesian classifier (BYC) decoding approach computes the posterior probability of a given movement state directly from extracted features, providing a principled probabilistic estimate of MI intent [190]. These CVA-derived features are often paired with a Gaussian classifier (GSC), which models each MI class as a Gaussian distribution in feature space and assigns labels based on maximum likelihood [95]. Moreover, the Markov switching model (MSM) models MI EEG signals as transitions between latent brain states governed by a Markov process, allowing the classifier to capture temporal structure and state-dependent feature dynamics that reflect how MI patterns evolve over time [236], [237].

2) Unsupervised Learning

Unsupervised learning algorithms, such as clustering or representation-learning methods, attempt to group MI EEG patterns based on their intrinsic structure without using class labels. While they can reveal latent patterns or assist with feature extraction, their direct application to MI classification is relatively limited compared to supervised approaches, which typically achieve higher accuracy by leveraging labeled MI data [22], [238].

3) Artificial Neural Networks

Artificial neural networks (ANNs) are inspired by the structure and function of biological neurons in the human brain, as depicted in Fig. 18 (a). A natural neuron receives electrical impulses through dendrites, processes them within the cell body, and transmits responses via the axon through synapses to other neurons. Similarly, an artificial neuron receives input signals, processes them mathematically, and sends output to connected units, emulating how electrical signals propagate through biological networks to enable learning and information transfer [239], [240]. An artificial neuron, as shown in Fig. 18 (b), consists of inputs, adjustable weights, a bias term, and an activation function. Each input is multiplied by its corresponding weight, summed up with the bias (the multiply–accumulate step), and then passed through an activation function that introduces nonlinearity. Common activation functions include Sigmoid and tanh (for smooth transitions), ReLU and Leaky ReLU (for efficient gradient propagation), Maxout, and ELU (for improved robustness), producing the final output that feeds into subsequent network layers. Typical ANN-based MI features classifiers are grouped and discussed in the following categories [241].

A) Traditional & Multilayer Models: The feedforward neural networks (FFNNs) process information in a strictly forward direction, moving from input to output through one or

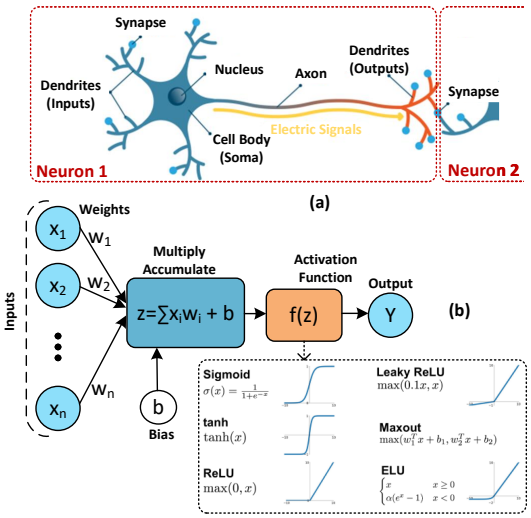

Fig. 18. (a) Natural biological neurons structures. (b) Symbolic mathematical representation of conventional artificial neurons.

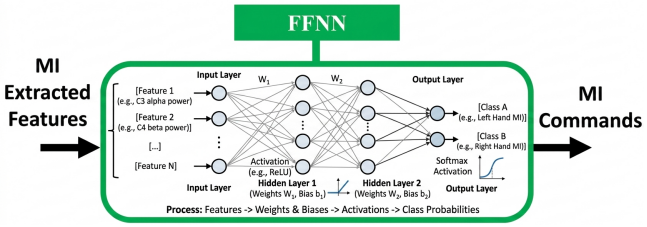

Fig. 19. FFNN-type ANNs used as MI feature classifiers.

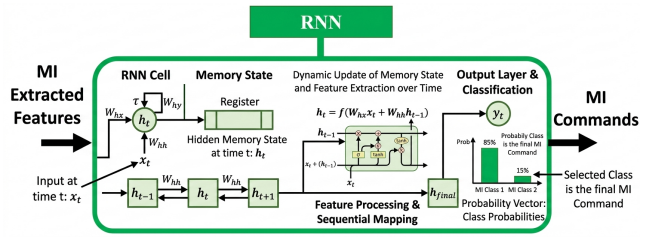

Fig. 20. The RNN-type ANNs as the MI features classifier.

more hidden layers without any feedback loops [142]. Each neuron performs a weighted sum of its inputs, adds a bias, and passes the result through a nonlinear activation function, enabling the network to learn complex decision boundaries [82]. The standard neural networks and multilayer Perceptrons (MLPs) extend this structure by stacking multiple hidden layers, allowing the model to capture hierarchical feature representations [211]. The FFNNs trained with backpropagation (BPNN) iteratively adjust weights based on error gradients to improve CA [55], [241]. In MI EEG classification, these models serve as powerful discriminators by learning nonlinear mappings from extracted MI features, such as time-frequency or spatial patterns, to class labels, offering flexible and effective pattern recognition when sufficient training data and well-engineered features are available.

FFNN-type ANNs used as the MI features classifier can be described in few steps, as shown in Fig. 19. (1) Input Layer: The extracted MI feature vector—comprising elements such as C3 alpha power, C4 beta power, and other numerical signal indicators $[Feature\,1, Feature\,2, \ldots, Feature\,N]$—is fed into the network. (2) Hidden Layers (Layers 1 and 2): Information moves exclusively forward through fully connected nodes, where inputs are multiplied by adaptive weights (W_1, W_2), combined with biases (b_1, b_2), and mapped via nonlinear activation functions (e.g., ReLU) to resolve intricate cross-feature dependencies. (3) Output Layer: The final layer applies a Softmax activation function to compute distinct class probabilities (e.g., Left Hand vs. Right Hand MI), translating the continuous internal feature maps into a definitive, discrete MI command.

B) Recurrent Neural Networks: The recurrent neural networks (RNNs) are designed to model temporal dependencies in sequential data by incorporating feedback connections that allow information from previous time steps to influence current processing [167], [184]. This structure enables them to capture dynamic patterns and context over time, an essential property for analyzing MI EEG signals, which unfold sequentially. Variants of RNNs enhance this ability: the echo state network (ESN) uses a large, fixed recurrent reservoir with trainable output weights for efficient temporal pattern learning; the long short-term memory (LSTM) network introduces memory cells and gating mechanisms to preserve long-range dependencies and mitigate vanishing gradients [146], [228]; the gated recurrent unit (GRU) simplifies the LSTM structure while maintaining effective sequence modeling [167]; and HYB CNN–LSTM architectures combine convolutional layers for spatial feature extraction with LSTM layers for capturing temporal evolution, achieving robust MI feature classification by learning both spatial and temporal characteristics of EEG data [148], [162].

RNN-type ANNs perform MI features classification in few steps, as shown in Fig. 20. (1) Sequential Input Tracking: The input MI extracted features are treated as a time-series sequence, where x_t represents the incoming feature slice at time step t. (2) Hidden Memory State Dynamics: An internal recurrence loop tracks sequential context through an adaptive hidden memory state (h_t). This memory state is dynamically updated over time via the mapping function $h_t = f(W_{hx}x_t + W_{hh}h_{t-1})$, effectively capturing temporal dependencies across successive motor imagery frames. (3) Output Layer and Sequential Mapping: The final cumulative sequence state (h_{final}) is fed into the output layer to compute a class probability vector (e.g., 85% probability for MI Class 1 vs. 15% for MI Class 2), routing the highest-probability index to generate the final discrete MI command.

C) Fuzzy-Integrated Neural Models: The fuzzy neural networks (FNNs) combine the learning capability of neural networks with the reasoning mechanism of fuzzy logic to handle uncertainty and imprecision in data. They operate by first transforming input features into fuzzy membership values (fuzzification), applying fuzzy inference rules to model relationships between inputs and outputs, and then producing a final decision through defuzzification. The neural network component adjusts membership functions and rule parameters

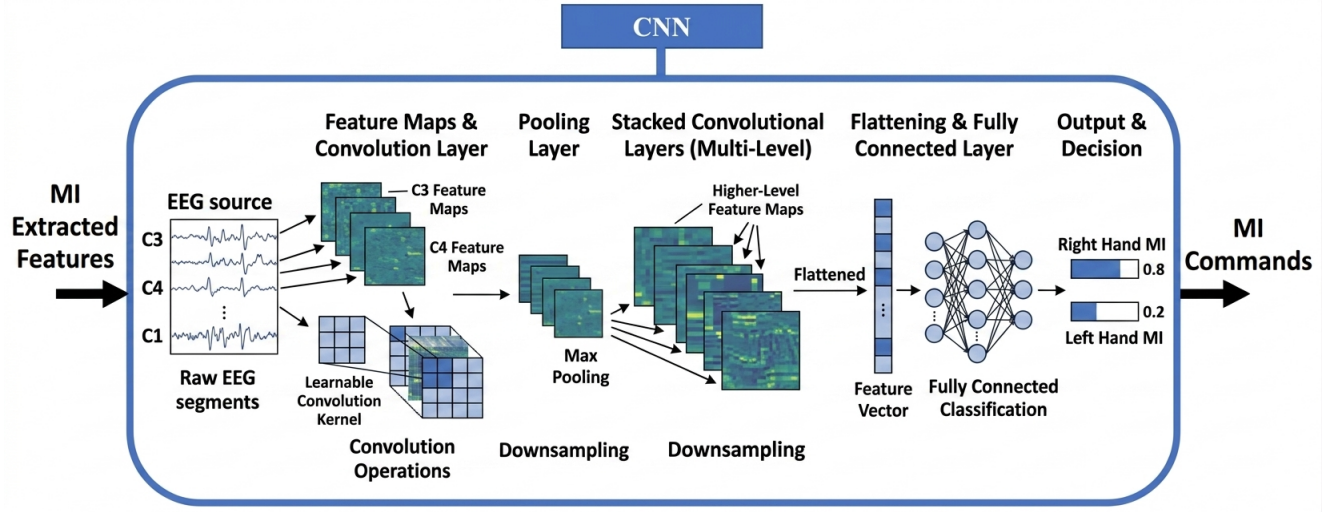

Fig. 21. Utilizing the CNN-based model for MI features classification.

through learning algorithms, typically gradient-based optimization. In MI EEG classification, FNNs are particularly useful because EEG signals are noisy and highly variable; the fuzzy framework effectively models ambiguity in MI features (e.g., spectral or spatial patterns), while adaptive learning improves CA and robustness across subjects [224], [242].

D) Probabilistic Neural Models: The probabilistic neural models apply statistical decision principles to classify input data based on estimated probability distributions rather than deterministic boundaries. In probabilistic neural networks (PNNs), classification is achieved by computing likelihood functions for each class using kernel-based estimations, and selecting the class with the highest posterior probability; this makes them inherently robust to noise and well-suited to small training sets [226], [243]. Although tree-based gradient boosting models such as XGBoost are not truly neural networks, they share a multilayer decision flow concept, aggregating many weak learners (decision trees) into a strong predictive model through iterative boosting. For MI feature classification, both PNNs and XGBoost offer complementary advantages: PNNs provide smooth probabilistic modeling of EEG patterns, while XGBoost efficiently captures nonlinear relationships and feature interactions, resulting in high accuracy and generalization across subjects and sessions [244], [245].

4) Deep Learning The DL algorithms are increasingly used for MI features classification due to their ability to automatically learn hierarchical feature representations directly from raw or minimally processed EEG signals [135], [159]. Unlike traditional handcrafted feature extraction methods, deep models can capture complex spatial–temporal patterns and inter-channel dependencies in EEG data [98], [165]. This leads to improved CA, better generalization across subjects, and reduced reliance on expert feature design, making DL a powerful and adaptive approach for modern MI systems. The most common DL-based classification algorithms can be categorized as follows.

A) Convolutional Neural Network: The CNNs operate on the principle of hierarchical feature extraction through convolutional filters that learn spatial or temporal patterns from input data [33], [131], [135], [145], [170], [173]. In MI EEG classification, 1D CNNs typically process temporal signal patterns from individual EEG channels, while 2D CNNs treat multi-channel EEG features (e.g., time–frequency maps) as images to capture spatial correlations across brain regions [140], [158], [172]. The multi-scale CNNs (MSCNNs) further enhance performance by applying filters of different sizes to extract features at multiple temporal or spatial resolutions, helping the model learn both global and local EEG dynamics. Also, VGG-16 has occasionally been adopted through transfer learning for EEG image-based classification despite its relatively high computational complexity [164]. Furthermore, hybrid-scale CNN (HS-CNN) employs convolution kernels of multiple sizes together with data augmentation to capture EEG features at different temporal scales and improve classification robustness [156].

Network design choices also range from monolithic CNNs, which integrate all processing stages within a single end-to-end architecture, to modular networks that decompose feature extraction and classification into separately trained, interchangeable sub-networks [198]. Architectural variants such as spatial CNNs, which emphasize inter-channel spatial filtering, and standard CNNs, which apply conventional convolutional stacks without specialized spatial or temporal branches, have also been benchmarked for MI decoding, often alongside classical baselines such as SVM with a cubic polynomial kernel [200]. The EEGNet, a compact and efficient CNN architecture designed specifically for BCI applications, uses depthwise and separable convolutions to learn frequency-specific activations while maintaining a lightweight design suitable for real-time systems [191]. Additionally, pre-trained CNNs such as AlexNet, ResNet50, and InceptionV3, originally trained on large visual datasets, can be fine-tuned on EEG-derived image-like features (e.g., spectrograms), leveraging transfer learning to improve accuracy and reduce training time. The model-agnostic meta-learning (MAML) and MAML-based lightweight neural networks (MAML-LNNs) have recently

been proposed to improve subject adaptation under limited calibration data by learning initialization parameters that rapidly adapt to new users [143]. Additionally, the compact temporal convolutional network (CTCNet) reduces computational complexity for online MI decoding while preserving high CA, making it well suited for real-time BCI operation [157], [246]. Moreover, lightweight ConvNeXt models with a Softmax classifier have also been developed to enable efficient subject-independent MI classification with reduced computational complexity [138].

Utilizing the CNN-based model for MI features classification includes a few steps, as depicted in Fig. 21. (1) Feature Maps and Convolution Layer: Input raw EEG segments from target spatial channels (e.g., C3, C4, C1) are scanned by learnable sliding convolution kernels to extract localized spatial-temporal feature maps. (2) Pooling and Multi-Level Downsampling: Max pooling layers downsample the generated matrices, reducing spatial dimensionality while retaining invariant structural features before passing them through stacked multi-level convolutional layers to form deeper high-level feature maps. (3) Flattening and Fully Connected Layer (FCL): The final multi-dimensional feature maps are flattened into a single-column 1D feature vector and routed into a fully connected classification network. (4) Output and Decision: The network processes the integrated features to compute final class probabilities (e.g., 0.8 for Right Hand MI vs. 0.2 for Left Hand MI), generating the corresponding definitive MI command.

B) Hybrid CNN–Recurrent Models: The HYB CNN–recurrent models combine the strengths of convolutional and sequence-based architectures to improve MI EEG classification by jointly learning spatial and temporal dynamics [228]. In these models, CNN layers first extract local spatial or time–frequency features from raw EEG signals or transformed representations, capturing patterns such as sensorimotor rhythms or localized activations. The extracted feature maps are then passed to LSTM layers, which model temporal dependencies, sequential fluctuations, and long-range correlations inherent in MI tasks [148], [167]. This HYB structure enables the system to learn both where and when discriminative patterns occur in the EEG, making CNN-LSTM architectures particularly effective for decoding complex MI-related brain activity [152]. Furthermore, CNN-LSTM feature fusion network (FFN) extracts spatial features using CNNs and temporal features using parallel LSTM modules before fusing intermediate representations for classification [148]. Additionally, hybrid deep neural network with transfer learning (HDNN-TL) combines CNN and LSTM networks with transfer learning to learn shared spatial-temporal representations while adapting efficiently to new subjects [152].

C) Deep Belief Network: The deep belief networks (DBNs) are hierarchical generative models composed of stacked restricted Boltzmann machines (RBMs) that learn to capture complex, high-order statistical dependencies within data [247]–[249]. As MI feature classifiers, DBNs automatically extract meaningful patterns from EEG signals by learning unsupervised representations at multiple abstraction levels, reducing noise and highlighting discriminative features relevant to different MI tasks. The convolutional deep belief network (CDBN) extends this concept by introducing convolutional and pooling operations, enabling localized and translation-invariant feature extraction in EEG time–frequency or spatial domains [182]. Meanwhile, the frequential deep belief network (FDBN) focuses on learning spectral features directly from EEG frequency bands, improving sensitivity to oscillatory components such as μ and β rhythms essential for MI decoding [177]. These architectures allow robust, data-driven feature learning that supports high CA even with limited labeled EEG data.

D) Transformer-Based Models: The DL-based transformer (TRF) architectures, the foundational models behind modern LLMs such as ChatGPT, have emerged as powerful MI feature classifiers [144], [250]. They utilize self-attention mechanisms to capture complex dependencies within EEG signals while eliminating the need for recurrent structures [134], [145]. In MI decoding, TRs treat EEG time samples (or short time segments, channels, or time–frequency patches) as a sequence of tokens and use multi-head self-attention to learn how different time points, channels, and frequencies relate to each other [53], [197], [250]. This allows the model to emphasize the most informative neural patterns, such as task-specific modulations of μ and β rhythms, while down-weighting noisy or irrelevant activity. Positional encodings preserve temporal order, enabling the TR to capture both short- and long-range temporal dynamics critical for MI tasks. As a result, TR-based models can learn rich, global spatio-temporal representations of EEG data, often achieve competitive or superior classification performance, and offer better scalability to longer recordings and multi-channel setups [251], [252]. Furthermore, temporal-spatial transformer (TST) employs self-attention across both temporal and spatial EEG dimensions to capture global dependencies and improve MI classification performance [134].

TR-based models used as MI feature classifiers perform classification in a few steps, as shown in Fig. 22. (1) Input Embedding and Positional Encoding: The incoming MI extracted features are partitioned into sequential feature snippets $[f_{\text{time}1}, f_{\text{time}2}, \ldots, f_{\text{time}n}]$; these snippets are projected via an embedding layer and combined with positional encodings (Pos_n) to preserve critical temporal sequence information. (2) Transformer Encoder Block: The encoded sequences pass through a Multi-Head Self-Attention layer—mapping Queries (Q), Keys (K), and Values (V) via scaled dot-product attention to capture long-range global and local dependencies—followed by residual connections (Σ), layer normalization (Add & Norm), and a position-wise Feed-Forward Network (FFN). (3) Flattening and Linear Head: The multi-head contextual representations are flattened into a unified classification head to extract integrated spatial-temporal profiles. (4) Classification Probabilities: The network evaluates the final vectors to yield distinct target command probabilities (e.g., Class A/Left Hand vs. Class B/Right Hand), which feed into decision logic to trigger the final discrete MI command.

5) Generative AI (GenAI): The generative AI (GenAI) models are built on the principle of learning data distributions rather than just discriminative boundaries. They aim to capture

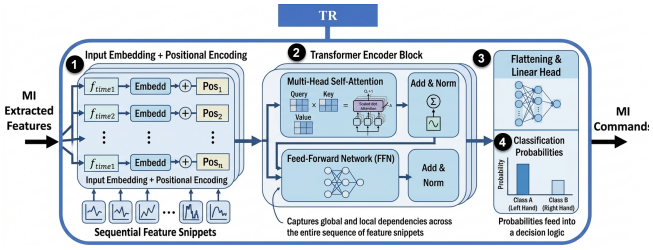

Fig. 22. The TR-based models as the MI features classifier.

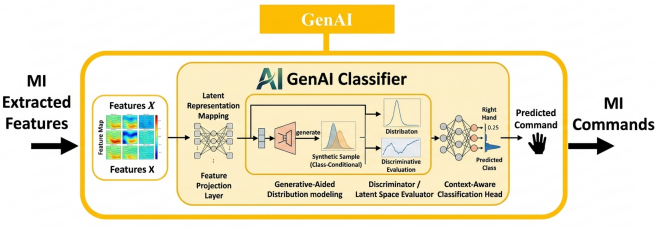

Fig. 23. The GenAI-based models as the MI features classifier.

and reproduce the underlying structure of the input data, enabling the generation or reconstruction of new, realistic samples. Among these, stacked autoencoders (SAEs) and variational autoencoders (VAEs) are prominent architectures [146], [235]. The SAEs consist of multiple layers of encoders and decoders that learn hierarchical representations by minimizing reconstruction error, allowing them to extract compact, non-linear features from EEG data that reveal spatial and temporal patterns useful for MI classification [175]. Moreover, the VAEs, as probabilistic extensions of autoencoders, learn latent variable distributions and enable smooth interpolation between feature states, effectively capturing the variability and complexity of EEG signals [100]. For MI decoding, these generative frameworks can uncover deep latent representations of neural activity, improving robustness against artifacts and enhancing generalization across subjects or sessions.

The performance of GenAI-based models as the MI features classifier is shown in a few steps, as shown in Fig. 23. (1) Feature Projection and Latent Mapping: The multi-dimensional input MI extracted features (X) are routed through a projection layer to map the signal topologies into a continuous latent representation space. (2) Generative-Aided Distribution Modeling: Class-conditional generative components (such as diffusion, VAE, or GAN architectures) model the underlying feature distributions to construct high-fidelity synthetic samples and establish robust density patterns. (3) Discriminator and Latent Space Evaluation: A discriminative evaluator assesses both raw and generated feature distributions to extract fine-grained boundary probabilities. (4) Context-Aware Classification Head: The evaluated latent features are passed to a deep classification network to compute final class metrics (e.g., Right Hand vs. Left Hand MI probabilities), yielding the predicted intent to output the final discrete MI command.

6) Fuzzy Models Fuzzy models classify MI EEG signals by handling uncertainty through degrees of membership instead of fixed boundaries [253]. They use adaptive "if–then" rules and membership functions to represent nonlinear, overlapping brain patterns [224]. This approach captures subtle EEG variations, offering both robustness and interpretability in MI feature classification. The typical fuzzy models used for MI classification can be grouped as follows.

A) Type-1 Fuzzy Logic Systems: Type-1 fuzzy logic–based classifiers use fixed membership functions to handle uncertainty in MI EEG signals by mapping features to classes through interpretable rules. The classical fuzzy logic (FLO) and general fuzzy rule systems define "if–then" relationships, while Takagi–Sugeno–Kang fuzzy neural network (TSK FNNs) combine fuzzy rules with neural networks for adaptive, data-driven learning. Also, the triangular fuzzy numbers (TFNs) provide simple, efficient membership shapes, allowing these models to capture nonlinear MI patterns with clarity and low computational cost [31], [168]..

B) Type-2 Fuzzy Logic Systems: Type-2 FLO systems extend classical fuzzy models by introducing uncertainty into the membership functions themselves, enabling more flexible handling of EEG variability. The interval type-2 FLO systems (IT2FLS) model this uncertainty using upper and lower membership bounds, while IT2-TFNs represent it with interval-based triangular fuzzy sets. Also, SRIT2NFIS combines this higher-order fuzzy reasoning with neural adaptation, offering enhanced robustness and generalization for complex, noisy MI feature classification [31], [217].

C) Neuro-Fuzzy Systems: Neuro-fuzzy systems (NFS) combine fuzzy logic's interpretable rules with neural networks' learning ability to classify MI features adaptively from EEG data [224]. Models like FNN and adaptive neuro-fuzzy inference system (ANFIS) learn fuzzy rules and membership parameters directly from data, while optimization-enhanced schemes such as fuzzy system optimized using PSO (FZ-PSO) refine fuzzy structures using evolutionary search [224]. Furthermore, self-regulating interval type-2 neuro-fuzzy inference system (SRIT2NFIS) further extends this by integrating interval Type-2 fuzzy sets into a neuro-fuzzy inference architecture, improving robustness to noise and uncertainty in MI classification [254], [255].

D) Fuzzy Integral-Based Models: The fuzzy integral (FZI) model fuses multiple MI features or classifier outputs by modeling their interactions through fuzzy measures rather than assuming independence [253]. This allows the system to capture nonlinear relationships and complementary information among EEG features, leading to improved decision reliability and robustness in MI classification tasks. A related approach, multiple linear discriminant analysis with fuzzy integral fusion (MLDA-FZI), first applies MLDA to classify EEG patterns within each sub-band spectrum independently, then fuses the sub-band decisions using a fuzzy integral to produce a final, more robust MI classification [223]. Other intelligent classifiers include the Self-Regulated Supervised Gaussian Fuzzy Adaptive System (SRSG-FasArt) [171].

7) Learning Paradigms & Hybrids

The HYB learning paradigms for MI EEG classification integrate complementary model types to better capture spatial, temporal, and subject-specific variability in brain signals [149], [151]. Architectures such as CNN–LSTM, CNN-TRF,

and RNN–LSTM combine convolutional or recurrent feature extractors with LSTM layers to jointly learn spatial patterns and long-range temporal dependencies [148]. Transfer learning (TL) with CNN-based or EEGNet-style models reuses knowledge from related tasks or subjects, reducing training data demands and improving generalization [161], [191], [220]. The supervised self-training (SST) strategy further reduces calibration burden by progressively fine-tuning a pre-trained subject-independent model using labeled online data, enabling calibration-free BCI operation without a dedicated offline session. The continual learning (CNL) mechanisms allow the classifier to incrementally incorporate new MI patterns while retaining previously learned knowledge, mitigating the performance degradation common in extended MI-BCI sessions. Ensemble techniques further boost robustness by aggregating diverse base classifiers, smoothing out individual model errors [100].

On the other hand, the optimization-driven HYBs enhance these pipelines by automatically tuning model structures and decision mechanisms. The ESN coupled with GTA, FI optimized via PSO, and Multi-objective Grey Wolf optimization for twin SVMs all exemplify search-based calibration of hyperparameters, feature weights, or fusion rules [212]. The HYB fuzzy–neural and fuzzy–optimization approaches, along with HYB CNN or RNN-DL pipelines, exploit the interpretability of fuzzy logic, the representational power of deep networks, and the exploratory strength of metaheuristics to yield more accurate and robust MI feature classifiers under noisy and non-stationary EEG conditions. Also, the weighted sum of prediction scores (WSPS) is a simple decision-level fusion strategy that combines outputs from multiple classifiers using weighted confidence scores [53].

8) Other Classification Methods Finally, a wide range of additional MI classification techniques falls into an "other methods" category, encompassing approaches that do not fit neatly into conventional model families. These include statistical classifiers, signal-quality assessment frameworks, and functional-connectivity-based analyses that capture relationships between EEG channels rather than relying solely on localized features. Various alternative MI detection strategies, ranging from correlation-based metrics to graph-theoretic models and traditional pattern-recognition schemes, also contribute valuable perspectives, offering complementary ways to improve MI decoding performance [150].

E. EEG Hardware-Implemented Algorithms

Practical implementation of EEG-based MI classification systems on hardware platforms is a crucial step toward translating research algorithms into real-world applications [35], [256]. Hardware deployment not only validates an algorithm's practical feasibility but also reveals its robustness under realistic operating conditions, including noise, non-stationary signals, and limited computational resources [257]. Furthermore, implementing EEG-based MI classifiers on physical platforms ensures that these systems can perform reliably in everyday scenarios, bridging the gap between laboratory research and functional, real-time MI-BCI solutions. A list of the most important hardware implementations of MI systems is reported in Table V. In comparison to Table IV, in this table, a new hardware (HW.) column is identifies the hardware platform utilized for the practical implementation. Also, the following keywords are considered in this table: resource usage (REU), power consumption (PC), area efficiency (ARE), speed-up (SPU), latency (LAT), memory reduction (MMR), inference time (INT), reconstruction accuracy (RA), and compression ratio (CR).

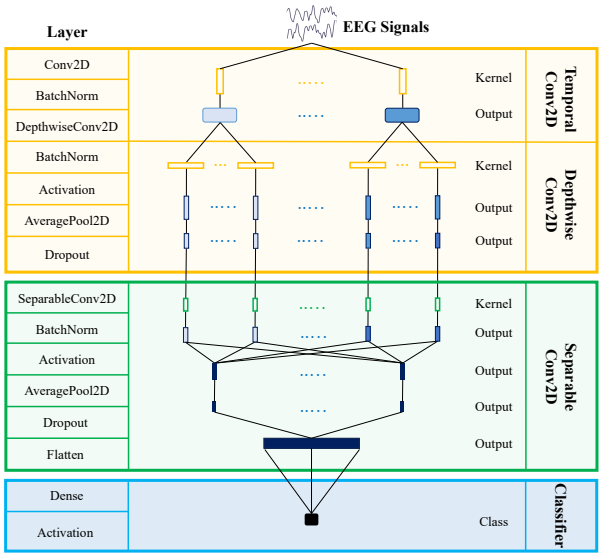

Fig. 24. The EEGNet CNN-based processing architucture for MI signals classification.

The state-of-the-art algorithms for EEG-based MI systems are evaluated primarily through rigorous benchmarking and a set of critical performance metrics, including CA, latency, robustness against noise, generalization across subjects, and computational efficiency. Since EEG signals are inherently non-stationary and noisy, software algorithms must extract discriminative temporal–spatial features while maintaining low complexity to support real-time operation. Furthermore, model compactness and decision interpretability are also essential factors for deployment in wearable or portable systems [14], [259].

In contrast, hardware benchmarking for MI-BCI systems focuses on the physical implementation and operational efficiency of the platform that executes these algorithms. Hardware evaluation emphasizes metrics such as real-time processing capability, power consumption, memory utilization, throughput, latency at the system level, and silicon area or resource usage (e.g., on FPGA or ASIC) [232], [260]. While software benchmarks primarily measure algorithmic performance and predictive capability, hardware benchmarks assess how efficiently those algorithms can be implemented and executed on embedded or edge devices [206], [261]. The key difference is therefore that software evaluation focuses on algorithm accuracy and robustness, whereas hardware benchmarking concentrates on implementation efficiency and system-level constraints required for practical, low-power, real-time BCI deployment.

TABLE V
STATE-OF-THE-ART EEG-BASED MI-BCI HARDWARE IMPLEMENTATIONS REPORTED IN THE LITERATURE IN RECENT DECADE.

Ref.	Year	HW.	Pre-processing	Feature Extraction	Feature Classification	Dataset Category	Performance
[30]	2026	Wearable EEG headband, 5 electrodes (4 wet, 1 dry)	NFL, BPF, BWF, CAR	CTCNet	CTCNet, SST	(8)	SICA: 86%
[197]	2026	wireless EEG system (InMex; WellFulfill Ltd., Taoyuan, Taiwan); Bluetooth; 8 dry electrodes; Oculus VR headset (paired with EEG cap for immersive AO+MI training)	BPF, BSF	CNN–TRF	CNN–TRF, CNL	(8)	CA: 47.4% (Session 1), 66.2% (Session 3); 52.8% (offline cross-validation benchmark)
[199]	2025	recoveriXPRO 16 channels stroke rehabilitation system; Estim FES device; 32-channel g.Nautilus EEG system; TMS device for motor evoked potentials (MEPs)/resting motor threshold (RMT)	BPF, BWF	LAP, PDC	LDA	(8)	Improved muscle strength in the wrist extensor (MRC-WE), MEAD:0.52, p:0.036
[14]	2022	60-channel EEG signal; Xilinx KC705 FPGA evaluation board; Artix-7 FPGA, Xilinx VC707 FPGA	BPF	EEGNet	EEGNet	(8)	CA: 93.06%; REU reduced to 2.54% area and 3.66% PC
[258]	2022	FPGA, 65 nm CMOS LP process	BPF	MC-EEGNet	MC-EEGNet	-	ARE: 87.22%, Eff: 20.77% better than a RISC-V MCU, SPU: 3.7×
[200]	2021	Wireless soft scalp electronics system with Samsung Gear VR; 6 flexible microneedle electrodes	BPF, BWF	PSD, WCM, CNN	Spatial CNN, Standard CNN, SVM (cubic polynomial kernel)	(8)	CA: 93.22 ± 1.33% (4-class, spatial CNN)
[201]	2020	FPGA Zybo board (Zynq xc7z010-clg400)	BPF	WPD	LDA	(10)	CA: 80%; Sen: 85%, Spe: 73.3%; AUC ≈ 0.79–0.80; ITR 2.78 bits/min; LAT: 7.5 ms; PC: 0.102 W.
[202]	2020	ARM Cortex-M4F MCU and ARM Cortex-M7	BPF, SWS	EEGNet	EEGNet	(6)	CA: 82.43% (2-class), 75.07% (3-class), 65.07% (4-class), LAT/energy: 101 ms & 4.28 mJ/inference (Cortex-M4F), 44 ms & 18.1 mJ/inference (Cortex-M7).
[203]	2020	Mr.Wolf PULP (Parallel Ultra-Low-Power) SoC	BPF	EEGNet CNN	Q-EEGNet, quantized EEGNet	(4)	SPU: 64×, MMR: up to 85%; 5.82 ms & 0.627 mJ per inference; 256× more Eff than EEGNet.
[57]	2020	Emotiv EPOC Flex (32-channel wireless EEG headset); MyoWare Muscle Sensor (EMG, AD8648 amplifier)	CAR, LAP, BPF	FFT	EUS	(8)	CA increased after MI training (e.g., 59–68% → 69–79%); overall ∼70% eff in modulating 7–12 Hz (mu) activity.
[198]	2019	OpenBCI device (8 dry electrodes, Ultracortex Mark IV headset)	BPF, Type II IIR filter, NFL, ITA	DFBCSP	Monolithic CNN, Modular network	(4), (7)	Dataset (4): Monolithic CA:78.41%, K=0.59. Modular: CA:80.03%, K=0.61. Dataset (7): Monolithic: CA:73%, K=0.64; Modular: CA:76%, K=0.67. INT: Monolithic 0.58s, Modular 1.04s per sample (GPU).
[205]	2019	FPGA — PYNQ-Z1 board (Xilinx Zynq-7020 chip)	AEP, HMW	CNN	CNN	(4)	CA: 80.5%, ∼7% CA drop with 8-bit fixed-point; subject a: 83%, subject b: 78%; FPGA 8× faster than computer. PC: FPGA 0.025 W vs. computer 140 W.
[192]	2019	Neurostyle nBETTER system EEG acquisition hardware; 24 unipolar Ag/AgCl electrodes (10-20 system), 256 Hz sampling, 24-bit resolution, ±300 mV range	BPF, EMD, CCA, LAP	FBCSP, MIN	FLDA	(8)	CA: ¿ 57.5%. A significant positive correlation between relative beta power (frontal/central) and CA.
[55]	2019	NeuroScan-NuAmps 40-channel EEG; Delsys 2-channel EMG	NFL, BPF, EMG processing	CSP, FFT	BPNN	(8)	CA: ¿80%; MAX joint angle tracking errors ¡0.1 rad.
[136]	2019	Virtex-6 FPGA, ML605 evaluation board	DC-blocker, LAP, CAR, BPF, HPF, NFL	SCSSP, MIN, LDA	LDA, SVM	(3),(4),(8)	CA: 73.54% (2, 3 classes); 67.2% (7, 4 classes); 80.55% (4, first 2 classes); 81.9% (9, 2 classes); LAT: 11.66 ms.
[206]	2018	Altera Stratix-IV FPGA development board	WOLA	CSP	LDA	(1),(2),(4),(8)	PC: 0.7 W, LAT: 0.430 s/trial. CA: 78.85% (4), 84.28% (1), 67.28% (2), 80.25% (8).
[211]	2018	Zedboard (Xilinx Zynq, ARM Cortex-A9 + FPGA fabric); Heterogeneous ARM+FPGA architecture; FPGA on Virtex-7 (XC7VX485T-2FFG1761C)	CPS	MLP	-	(8)	RA: 89.85% at 50% CR; CA: 78–93%; PSNR up to 23.6 dB.
[190]	2017	Custom BCI: 8-channel EEG amplifier; INA128 instrumentation amp + 2× OPA4241 op-amps. Conventional BCI (comparison): NeXus-32-channel commercial EEG amplifier.	BPF, NFL	LDA, PCA	BYC-Decoding	(8)	CA: 93.6±4.3% (custom), 96.2±1.8% (conventional).
[204]	2017	Custom real-time BCI system, 16 channels, UE-16B EEG amplifier	BPF, BWF	CSP, LCD, FFA, LAM, GTA, PSO	SRDA	(4),(8)	CA: 56.91% (4, SRDA alone), 70.2%, K=0.6 (4, FFA-LAM-SRDA).
[217]	2016	g.tec EEG amplifier with AgCl electrodes; 2 bipolar EEG channels over C3/C4 (10/20 system)	BPF	STFT, PCA	IT2FLS, LDA, FLDA, SVM	(11),(8)	Offline: IT2FLS significantly outperformed LDA, FLDA, and SVM (p_i0.05). Online: IT2FLS delivered statistically significantly higher CA than LDA,FLDA, and SVM.
[223]	2016	Custom wireless, portable EEG acquisition device; 5 dry electrodes, pre-amplifier, Bluetooth, Kinova robotic arm for real-time control output	FBD, HPF	SBCSP	MLDA-FZI-PSO	(8)	Offline (AUC, 4-fold CV ×10): Sugeno integral improved 0.968±0.063 → 0.998±0.040 after PSO; Choquet integral improved 0.992±0.014 → 0.998±0.003 after PSO (best overall); Online real-time test CA ≈86%.
[95]	2015	BCI-controlled telepresence robot; g.USBamp 16-channel active EEG electrodes amp., telepresence robot.	BPF, NFL, LAP, WCM PSD	CVA	GSC	(8)	CA: Online ¿70%. Time ratio (BCI/manual): 109.2±11.1% (end-users) vs. 115.1±10.3% (healthy), not significantly different (p_i0.1).

Modern DSP designs must balance high accuracy and robustness with low power consumption, real-time throughput, and hardware compactness. Achieving these optimization goals requires a holistic design approach, integrating algorithmic exploration with architectural and circuit-level innovations across both software and hardware domains. A wide range of hardware platforms, such as ASICs, FPGAs, DSP processors, and microcontroller-based systems running real-time operating systems (RTOS), play crucial roles in implementing efficient EEG-based MI-BCI systems [14], [202]. FPGAs offer high parallelism and reconfigurability, making them well-suited for prototyping and accelerating computationally intensive signal-processing and ML algorithms. DSP processors are optimized for real-time numerical operations such as filtering and feature extraction, making them effective for continuous EEG signal processing pipelines. RTOS-based microcontrollers enable deterministic task scheduling, low latency, and low power consumption, which are important for portable and embedded BCI devices [104]. ASICs, although less flexible, provide ultra-low power consumption, compact size, and very high processing efficiency, making them suitable for final deployment in wearable or implantable BCIs [260]. FPGAs are often preferred during research and development due to their flexibility and parallel processing capabilities, while ASICs

and low-power microcontrollers become more advantageous for large-scale, power-constrained MI-BCI deployments [259].

On the other hand, designing hardware-compatible algorithms is essential for deploying EEG-based MI-BCI systems in real-world environments. Classical algorithms such as CSP and SVMs are well-suited for hardware implementation because of their relatively low computational complexity and predictable memory requirements [55], [159], [206], [217]. More recently, advanced ML-based algorithms have enabled the deployment of hardware-friendly DL models, including quantized CNNs, lightweight RNNs, SNNs, and optimized architectures such as EEGNet and its low-precision variants [165], [197], [200], [205].

EEGNet is particularly attractive for hardware realization because it employs depthwise and separable convolutions, significantly reducing the number of trainable parameters and arithmetic operations while maintaining high classification performance for MI tasks [14], [203], as illustrated in Fig. 24. A model-compressed variant, MC-EEGNet, further reduces EEGNet's footprint through embedded channel selection, normalization merging, and product quantization, enabling more efficient on-device inference [258]. Similarly, Q-EEGNet applies quantization together with algorithmic and implementation-level optimizations to execute MI-BCI inference on resource-limited edge devices, removing the need for separate network reduction steps and generalizing readily to other CNN architectures [203].

The architecture first applies a temporal convolution layer to learn frequency-selective representations of MI-related EEG rhythms, followed by a depthwise convolution layer that performs efficient spatial filtering across EEG channels, and a separable convolution layer that further extracts discriminative spatio-temporal features with minimal computational cost. Batch normalization, ELU activation, average pooling, and dropout layers are incorporated throughout the network to improve training stability, reduce feature dimensionality, and prevent overfitting before the extracted features are passed to a fully connected Softmax classifier for final decision making. Owing to its lightweight architecture, EEGNet is highly suitable for FPGA, ASIC, and embedded edge implementations [14], [261], where techniques such as model compression, fixed-point arithmetic, quantization, and parallel pipelining further reduce hardware complexity, memory requirements, power consumption, and inference latency, enabling compact, energy-efficient, and real-time wearable BCI systems [14], [191], [202], [203]. Furthermore, the temporary constrained sparse group lasso (TCSGL-EEGNet) has recently been proposed to enhance EEGNet performance by enforcing structured sparsity, reducing computational complexity while maintaining CA [155].

F. General Overview of MI-BCI Signal Processing Pipeline Algorithms

As discussed in the previous sections, the signal processing pipeline for MI classification consists of several sequential stages, including preprocessing, feature extraction, feature reduction, and classification, each employing specific algorithms to transform raw neural signals into reliable control commands. In Fig. 25, the most common algorithms in recent literature over the past decade, as tabulated in Table IV and Table V, for each of these signal processing blocks of EEG-based MI-BCI are illustrated and categorized. Algorithms used in these stages span multiple methodological families such as classical signal processing techniques, supervised and unsupervised learning methods, DL models, GenAI approaches, fuzzy models, and evolutionary optimization algorithms [221]. As discussed previously, the feature extraction methods are commonly categorized according to the domain in which the EEG signal is analyzed, including frequency-domain techniques, time–frequency domain approaches, and spatial-domain methods, while dimensionality reduction techniques are used to select the most discriminative features and reduce computational complexity. In addition, filter bank techniques are widely employed to capture discriminative information across multiple frequency bands [141]. This broad spectrum of algorithms reflects the ongoing effort to improve MI-BCI performance, robustness, and computational efficiency by combining advances in signal processing and ML algorithms.

In the next section, some of the most promising technologies with the potential to revolutionize the current state of the art in MI-BCI systems are discussed.

VI. Converging Technologies for MI-BCI Systems

As illustrated in Fig. 26, the convergence of emerging hardware and software technological trends is poised to revolutionize EEG-based MI-BCI systems by transforming them from laboratory-bound prototypes into intelligent, wearable, and seamlessly integrated solutions. Advances in AI, wearable technology, system-on-chip (SoC) design, augmented and virtual reality (AR/VR), edge processing, hardware acceleration, energy harvesting, and the Internet of Things (IoT) collectively enable more adaptive, efficient, and user-centered BCIs [262]. Together, these technologies create a synergistic ecosystem that enhances real-time processing, mobility, and long-term usability, paving the way for next-generation MI-BCI systems that are both practical and pervasive in everyday life.

At the core of any MI-BCI system lies its signal-processing pipeline, which is now being transformed by the integration of powerful AI-based algorithms. Although early MI-BCI research in the 1970s primarily explored the basic feasibility of establishing direct neural communication pathways [1], [2], [263], recent decades have seen a major shift fueled by miniaturized, high-fidelity EEG sensing technologies paired with sophisticated computational techniques capable of interpreting non-stationary, low-SNR scalp signals [20], [24]–[27], [264], [265]. The performance of modern MI-BCIs is increasingly shaped by the combined evolution of high-precision software methods and improved hardware designs, where ML, DL, and HYB models have dramatically advanced the decoding of neural activity by enabling more reliable recognition of complex patterns, substantially boosting decoding performance and CA.

Wearable MI-BCI technologies should fulfill key design criteria such as portability, low-power consumption, wireless connectivity, ease of use for non-expert users, minimal setup time, high signal quality, robustness to motion artifacts, comfort during long-term wear, and cost-effectiveness [106], [266],

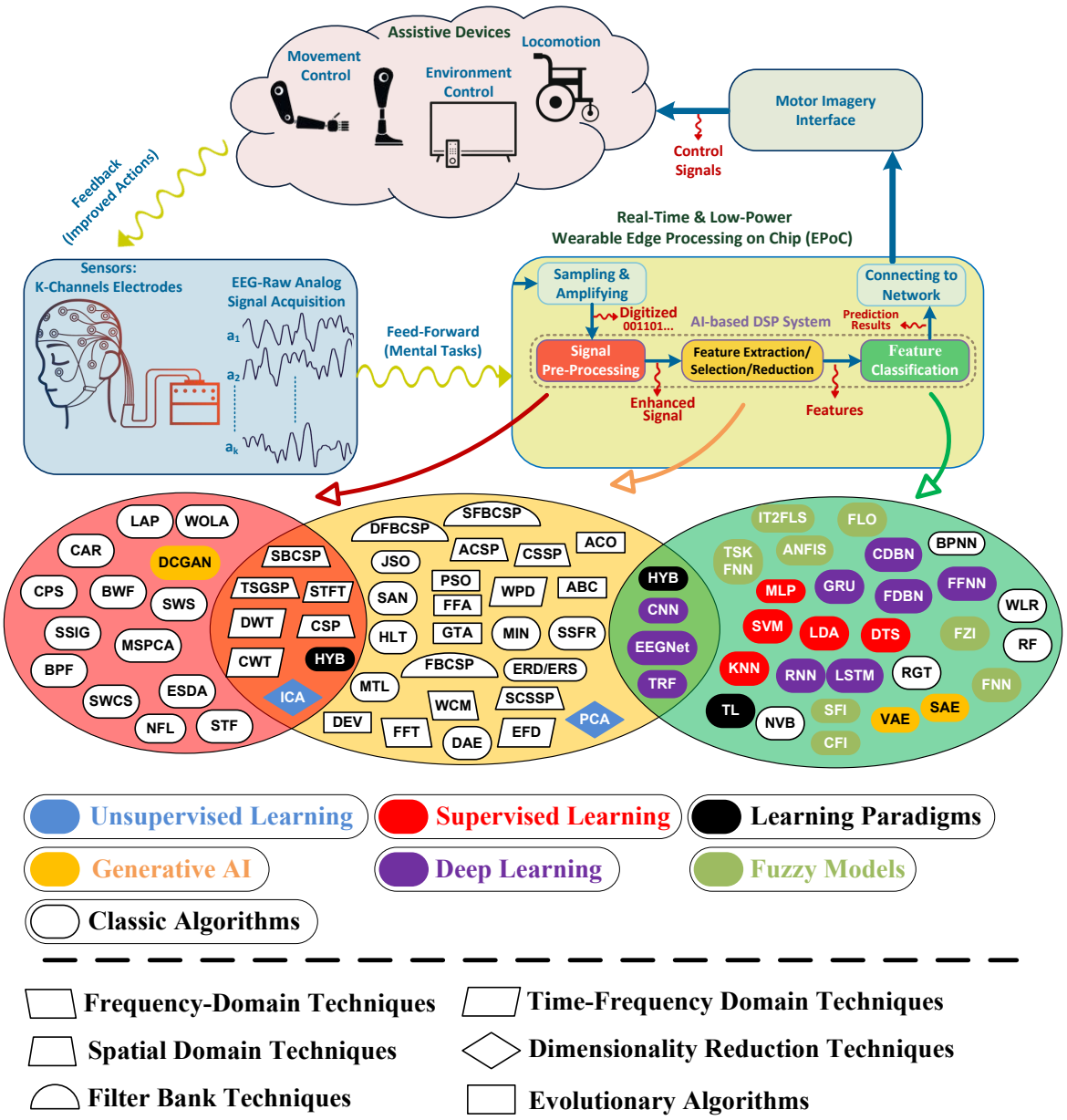

Fig. 25. State-of-the-art EEG-based MI-BCI system. The most common and most recent algorithms reported in the literature in recent decade for each blocks of the signal processing pipeline are illustrated.

[267]. Meeting these criteria ensures that EEG-based wearable systems can move beyond laboratory settings into everyday environments, enabling practical and continuous brain-computer interaction in real-life scenarios [97], [268].

EEG electrodes can be categorized by technology as dry, wet, and semi-dry, each offering distinct advantages and limitations. The dry electrodes provide fast setup and comfort without gels but often suffer from higher impedance and reduced signal quality. The wet electrodes, whether gel-based or saline-based, offer superior conductivity and lower movement artifacts but require time-consuming preparation, careful cleanup, and can cause user discomfort [213], [269]. The semi-dry electrodes strike a balance by using minimal electrolyte to improve signal conduction while maintaining an easier setup [270]. For instance, innovative flexible inkjet-printed electrodes [271] use silver nanoparticles to enable cost-effective, gel-free integration with mixed-signal SoC, demonstrating success in personalized healthcare applications. Moreover, emerging tattoo-like electrodes further enhance comfort and stability through ultra-thin, skin-conforming designs [272]. As these technologies advance, wearable EEG platforms significantly improve MI-BCI systems' usability by reducing cable-related artifacts, increasing mobility, guiding non-expert users in electrode placement, and enabling more portable, robust, and user-friendly systems suitable for real-life operation.

Some examples of different types of EEG electrodes are illustrated in Fig. 27. Dry electrodes utilize a multi-pin array designed to mechanically bypass the hair layer and establish direct metal-to-skin contact with the stratum corneum without

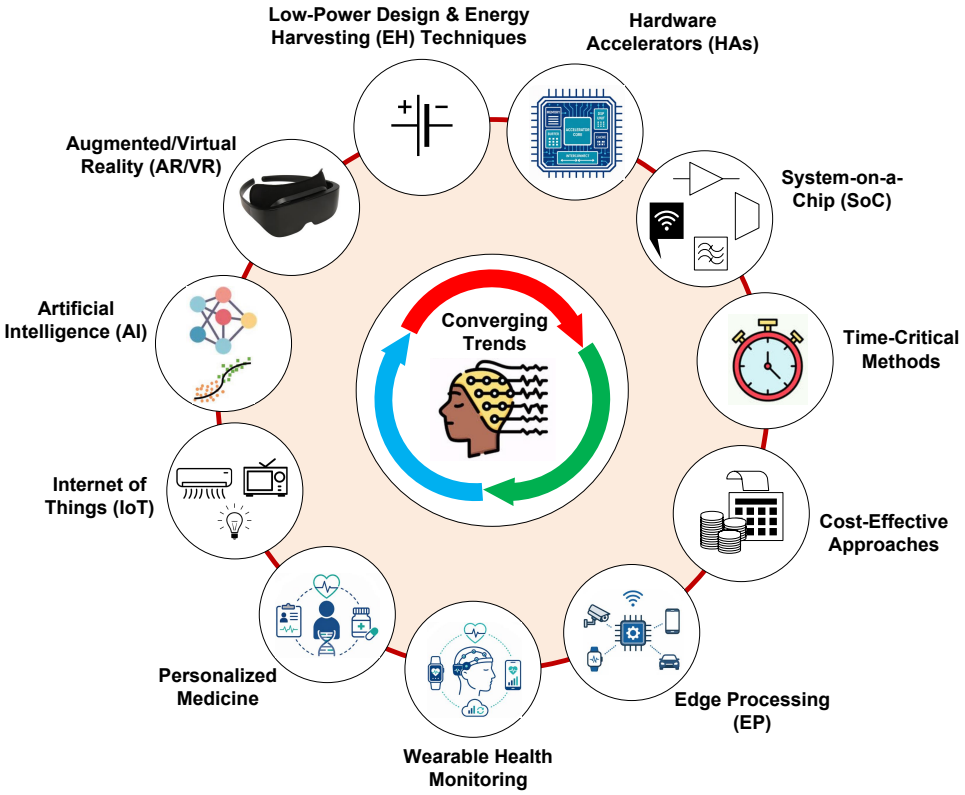

Fig. 26. Convergence of software and hardware technologies driving the commercialization of next-generation EEG-based MI-BCI applications.

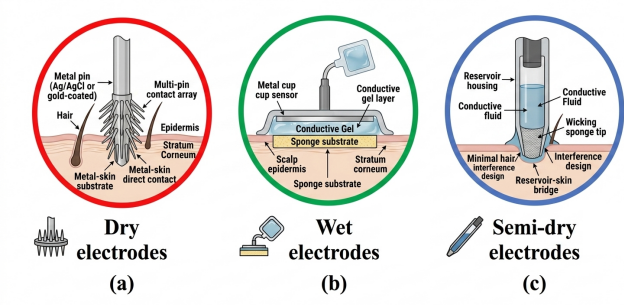

Fig. 27. Different types of EEG electrodes. (a) Dry electrodes. (b) Wet electrodes. (c) Semi-dry electrodes.

the need for conductive gel. Wet electrodes feature a traditional metal cup sensor paired with a sponge substrate or electrolyte chamber that contains a specialized conductive gel layer, lowering skin-electrode impedance by establishing a secure bridge to the scalp epidermis. The semi-dry electrodes employ a HYB reservoir housing that slowly releases a minimal amount of conductive fluid through a wicking sponge tip, combining a minimal hair-interference design with a continuous reservoir-to-skin liquid bridge to optimize preparation time and long-term signal stability [273], [274].

In EEG-based MI-BCI systems, the captured scalp signals can be transmitted to the processing unit either through traditional wired connections or via wireless technologies such as Bluetooth, proprietary low-power radio, or Wi-Fi [275], [276]. While wired EEG systems typically offer higher bandwidth and more stable data transmission, they require bulky helmets and numerous cables that prolong setup time and restrict mobility, making them unsuitable for everyday use. In contrast, modern wearable EEG devices integrate microchip electrodes with on-board amplifiers, quantizers, and wireless transmitters, enabling cable-free operation that is more comfortable, portable, and less prone to cable-movement artifacts. Bluetooth and similar low-power wireless protocols thus play a key role in realizing small, head-mounted systems that can stream MI-BCI data reliably over extended periods in real-life environments [106], [277], [278].

The integration of AR/VR into EEG-based MI-BCI systems has introduced powerful new avenues for neuro-rehabilitation and user interaction [279], [280]. By immersing users in realistic, closed-loop environments, AR/VR enhances engagement, motivation, and sensory feedback while enabling motor priming, an essential process for restoring or strengthening neural pathways [120], [280]. These immersive interfaces allow individuals to visualize and control virtual limbs or objects through thought-driven commands, thereby bridging the gap between neural intention and physical response [281]. As a result, AR/VR-enabled MI-BCI systems not only improve training outcomes and motor recovery but also create more intuitive and interactive experiences for both clinical rehabilitation and everyday neuroadaptive applications [282].

SoC designs, HAs, and edge processing (EP) are transforming EEG-based MI-BCI systems by tackling key challenges such as latency, size, and power consumption [257], [261], [283]. Integrating signal acquisition, preprocessing, and classi-

fication into compact SoC platforms enables miniaturized and less bulky wearable devices capable of recording EEG signals continuously for extended periods (days or even weeks). However, moving from high-power laboratory setups to real-world, resource-constrained environments remains a critical bottleneck. In this context, EP plays a central role by shifting computation closer to the sensors, thereby reducing latency, enhancing real-time responsiveness, and improving data privacy [284], [285]. To further support computationally intensive AI algorithms, dedicated HAs are increasingly employed to efficiently execute feature extraction and classification tasks while maintaining low power consumption.

Despite these advances, battery life continues to limit the widespread adoption of wearable MI-BCI systems outside controlled settings. EH techniques offer a promising solution by enabling autonomous and maintenance-free operation through the capture of ambient energy sources such as body heat or mechanical vibrations. When combined with low-power SoC architectures and optimized hardware accelerators, EH can significantly extend device lifetime and reduce reliance on frequent battery recharging [286], [287]. Overall, the convergence of SoC integration, edge computing, hardware acceleration, and sustainable energy solutions is paving the way toward fully wearable, intelligent, and long-term EEG-based MI-BCI systems suitable for real-world applications [283], [288].

The integration of IoT technologies into EEG-based MI systems extends their functionality beyond standalone operation toward interconnected, intelligent environments. By embedding BCI modules into IoT-enabled devices, these systems can continuously monitor neural signals and infer cognitive states such as mental fatigue, stress, attention, and frustration, enabling adaptive and context-aware responses in real time [289], [290]. This connectivity allows seamless data exchange with cloud or edge platforms for further analysis, remote monitoring, and personalized feedback, while also facilitating applications in smart healthcare, rehabilitation, and assistive living [114], [192]. Consequently, IoT-enhanced MI-BCI systems can act as intelligent assistants within smart environments, improving user experience and enabling more responsive and scalable neurotechnology solutions.

Emerging directions in EEG-based MI-BCI systems include the exploration of quantum computing, ultra-low-latency processing methods, and cost-effective design and fabrication strategies to enable broader adoption [291]. Although still in its infancy, the integration of quantum computing with BCI has introduced novel paradigms such as HYB classical–quantum frameworks and quantum brain networks, where neural signals can potentially interact with quantum systems (e.g., controlling or measuring qubits), opening new interdisciplinary research opportunities. At the same time, time-critical methods such as optimized signal pipelines, edge-based inference, and lightweight algorithms are essential to minimize latency and ensure real-time responsiveness, which is crucial for practical MI-BCI applications [292], [293]. Finally, for commercialization and widespread use, reducing system cost through scalable hardware, efficient architectures, and affordable wearable components remains a key priority, ensuring that these advanced neurotechnologies become accessible beyond specialized research environments.

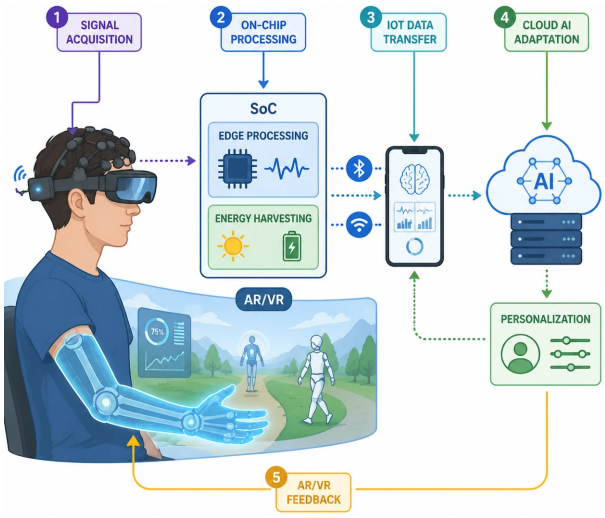

Fig. 28. Intelligent integrated MI-BCI ecosystem for personalized neuro interaction.

The utilization of several emerging technologies in an MI-BCI ecosystem is illustrated in Fig. 28 through an example of a system-level architecture that combines wearable sensing, EP, and cloud intelligence to enable mobile, adaptive BCI interaction. The user wears a lightweight wireless EEG headset equipped with flexible electrodes and an SoC that integrates amplification, ADC, and an EP core for real-time signal decoding. The SoC manages power through EH modules that capture body heat or motion, allowing low-power operation. Processed neural features are transmitted via Bluetooth or Wi-Fi, connecting the headset to nearby devices and the cloud in an IoT framework.

In the cloud–AI layer, accumulated data refines personalized neural models for improved movement decoding and adaptive performance. These personalized classifiers enable applications such as AR/VR control, neurorehabilitation, and cognitive training, where users interact with responsive virtual environments through MI. The figure illustrates how sensing, EP, wireless connectivity, and intelligent personalization converge into an energy-efficient, user-centered MI-BCI ecosystem, highlighting the synergy between HA, on-chip integration, and sustainable design principles.

In the next section, a modern AI-based EP SoC architecture is proposed for next-generation EEG-based MI-BCI systems. This architecture is designed to tightly integrate low-noise EEG acquisition, energy-efficient signal processing, and intelligent inference within a unified hardware platform, enabling real-time operation under strict power and latency constraints.

VII. CONCEPTUAL HARDWARE ARCHITECTURE FOR NEXT-GENERATION MI-BCI SYSTEMS

Based on the technologies and hardware implementations reviewed in the preceding sections, this section presents a conceptual SoC architecture for next-generation EEG-based MI-BCI systems. The proposed architecture is intended to illustrate how low-power EEG acquisition, on-chip signal

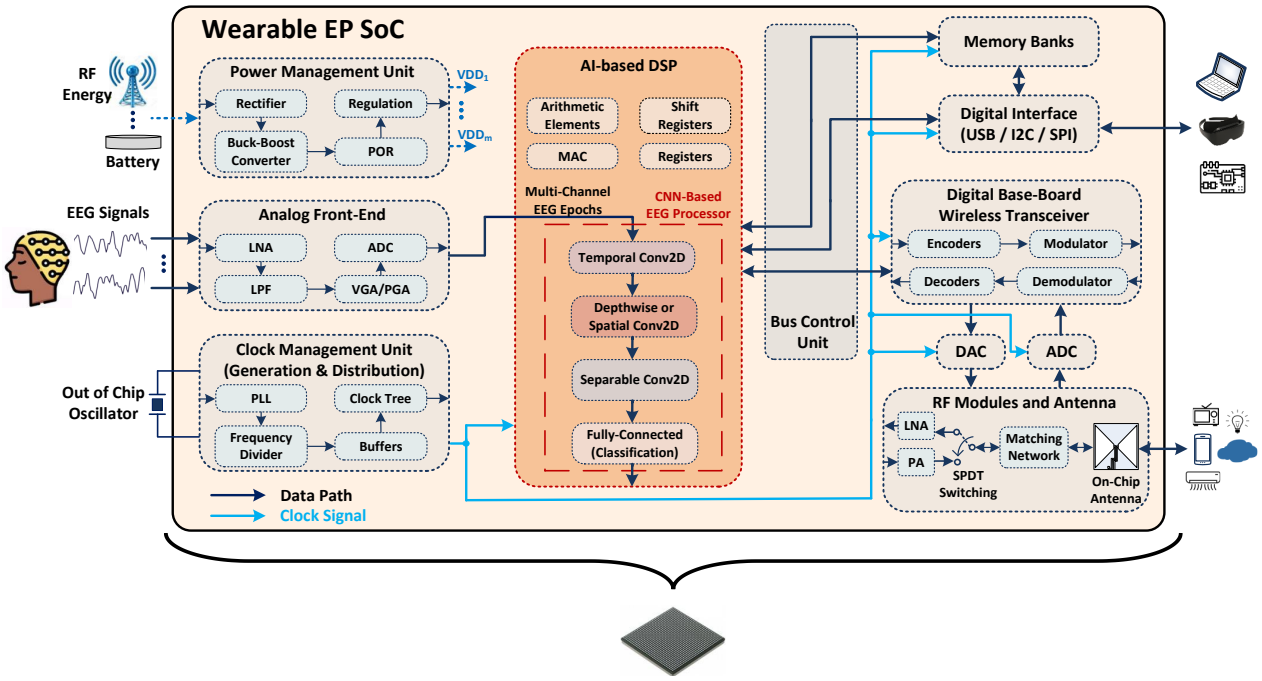

Fig. 29. Conceptual block diagram and major subsystems of a next-generation wearable edge-processing EEG-based MI-BCI SoC.

conditioning, AI-based signal processing, wireless communication, and power-management functions could be integrated within a unified platform. It should be emphasized that the proposed architecture is a conceptual system-level design and has not been physically implemented or fabricated.

The proposed block diagram of an AI-powered EEG-based EP MI-BCI chip is illustrated in Fig. 29. The conceptual SoC is intended to illustrate how efficient power management and real-time processing could be supported in future MI-BCI implementations. The primary features of this SoC include EH techniques to eliminate or reduce the need for a battery, integration with the IoT ecosystem through AI-enabled EP, and an optimized transceiver circuit for seamless wireless communication. These architectural features could potentially support a range of applications, such as medical signal recording for health monitoring and emergency alert systems. The proposed SoC comprises multiple interconnected sub-systems that collaboratively enable efficient and reliable processing of EEG signals for MI tasks.

At the core of this SoC is the AI-based DSP unit, designed to enhance EEG signal interpretation and classification using deep learning algorithms. This unit forms the computational backbone of the system and operates in a structured pipeline including pre-processing, feature extraction, classification, and post-processing. A CNN-based classification engine could be employed to support real-time MI prediction with potentially reduced processing latency, while a post-processing module could apply smoothing and multi-channel decision fusion to improve robustness.

The DSP architecture is inspired by EEGNet, a lightweight CNN model optimized for EEG-based BCIs. EEGNet employs temporal convolutions, depthwise spatial convolutions, and separable convolutions to efficiently capture both temporal dynamics and spatial correlations across channels while significantly reducing parameter count and power consumption. This compact yet powerful architecture has demonstrated strong performance on public datasets, outperforming traditional shallow and deep ConvNet models. Furthermore, its hardware-friendly structure makes it well-suited for real-time deployment on edge processors and ASIC-based EP platforms, where energy efficiency, speed, and classification reliability are critical.

The other sections of the proposed EP SoC are as follows:

A power-management unit (PMU), potentially integrated with energy-harvesting interfaces, could support low-power operation suitable for wearable applications. The PMU is intended to manage energy from multiple inputs, such as RF energy harvesters, micro-batteries, or other ambient sources, and to provide the voltage levels required by different system components. The rectifier first converts incoming RF or AC energy into DC voltage. The buck-boost converter then elevates and stabilizes this voltage to a usable level, buffering energy to handle load fluctuations. A power-on reset (POR) circuit ensures safe system startup by holding the SoC in reset until the supply reaches a defined threshold. Finally, the regulation stage provides stable, low-noise output voltages for the AFE and DSP blocks, ensuring consistent and efficient operation of the MI-BCI system.

The AFE unit filters, amplifies, and conditions the weak EEG signals before they reach the digital domain, ensuring that meaningful neural information is preserved with minimal distortion. It typically includes low-noise amplifiers (LNAs) to boost microvolt-level EEG signals while adding as little noise as possible, LPFs to remove high-frequency noise and

interference, and variable or programmable gain amplifiers (VGA/PGA) to adapt the amplification level based on signal amplitude and recording conditions. Together, these blocks maximize signal integrity and dynamic range, enabling the downstream ADC and processing engines to operate with higher accuracy and robustness.

The clock management unit is responsible for generating and distributing clock signals across the SoC. This sub-system includes phase-locked loops (PLL), frequency dividers, clock trees, and buffers to ensure precise timing and synchronization for efficient processing.

The memory-bank unit provides temporary storage for EEG signal processing, intermediate computations, and algorithm execution. These memory units support high-speed data access and real-time performance.

The digital interface unit enables communication with external components via universal serial bus (USB), inter-integrated circuit (I2C), and serial peripheral interface (SPI) interfaces, ensuring flexible integration with external devices and controllers. This interface also facilitates seamless connectivity with AR/VR platforms, allowing real-time transmission of decoded MI-BCI commands to head-mounted displays or immersive rehabilitation systems, thereby enabling interactive neurofeedback, virtual training environments, and low-latency brain-controlled user experiences.

The digital base-board wireless transceiver unit enables reliable wireless communication between the EEG-based MI-BCI system and external devices. In this section, encoders prepare outgoing data for transmission, the modulator maps it onto a carrier signal, the demodulator recovers the received signal, and the decoders reconstruct the original data, ensuring efficient and accurate bidirectional communication. Additionally, by integrating IoT technology, this transceiver supports seamless connectivity with cloud platforms and smart devices, enabling remote monitoring, data sharing, and adaptive learning for continuous improvement of user performance and device intelligence in real-world applications.

The wireless transceiver and RF modules handle digital signal transmission and reception, while RF modules and antennas manage radio frequency communication. The on-chip antenna, LNA, power amplifier (PA), and RF matching networks facilitate reliable wireless data exchange.

Finally, the bus control unit coordinates data flow between internal sub-systems, ensuring efficient and organized communication within the SoC. It also manages ADC and digital-to-analog converter (DAC) conversions, enabling seamless transitions between analog and digital signals.

In conclusion, the proposed wearable edge-processing EEG-based MI-BCI SoC represents a conceptual framework integrating AI-based DSP, power management, wireless connectivity, and on-chip computation within a unified architecture. Such integration could potentially reduce external data transmission, support low-latency processing, and enable more compact and energy-efficient future MI-BCI implementations. However, the proposed architecture has not been implemented or fabricated, and its power consumption, latency, memory requirements, silicon area, and system-level performance remain to be quantitatively evaluated in future work.

VIII. Discussions

After conducting an extensive and systematic exploration of the literature in the field of EEG-based MI-BCI systems, this section synthesizes the key findings and technical insights gathered from prior research. Building upon this foundation, we now proceed to address the research question outlined in Section II, connecting the surveyed evidence to the core objectives and motivations of this study.

RQ1. Which BCI application currently has the greatest practical impact and the strongest potential for future development and improving the quality of life? What is the role of EEG-based MI-BCI systems within this landscape?

To answer this question, a literature analysis was performed with a detailed review of titles, keywords, abstracts, and data description sections to ensure comprehensive coverage. Hence, the most popular and widely explored applications of EEG-based BCI technology are extracted and illustrated in Fig. 30(a). The figure presents both medical and non-medical applications, offering insights into the prevalence of each category. The percentages reflect the current research focus and potential areas for future advancements in the BCI domain. The most dominant application in EEG-based BCI is medical and assistive applications, accounting for 38.2%, highlighting its critical role in healthcare and rehabilitation. Other notable applications include: robotics and control (15.9%), cognitive enhancement and neurofeedback (11.1%), smart home and environment control (10.8%), neuromarketing and emotion recognition (9.4%), security and authentication (7.5%), and gaming and entertainment (7.1%).

The Fig. 30(b) provides a closer look at the prevalence of different medical applications of EEG-based BCI technology. These applications are crucial in assisting individuals with neurological conditions and improving healthcare solutions [99]. The primary medical applications include: MI (61.2%), neurological disorder diagnosis and monitoring (23.3%), sleep disorder diagnosis and management (10.7%), and chronic pain management (4.8%). Based on the literature, while assistive technology and rehabilitation remain the most widespread applications, the range of medical uses for EEG-based BCI has expanded significantly in recent years to include diagnosis and monitoring of various neurological and mental health conditions, as well as chronic pain management. However, some emerging applications like neurostimulation for depression and personalized interventions for autism are still in the early research stages.

Among the various medical applications of EEG-based BCIs, MI stands out as a crucial area of focus due to its significance in rehabilitation capability and its positive impact on improving human lives, enabling individuals with severe disabilities to control assistive devices such as wheelchairs, prosthetic limbs, and communication aids using brain signals. MI involves the mental simulation of movement without actual physical execution, allowing users to engage with BCI systems effectively. Case studies have highlighted the successful use of MI in enhancing recovery outcomes for individuals post-stroke or with spinal cord injuries [54]. For instance, patients have demonstrated improved motor function by practicing imagined

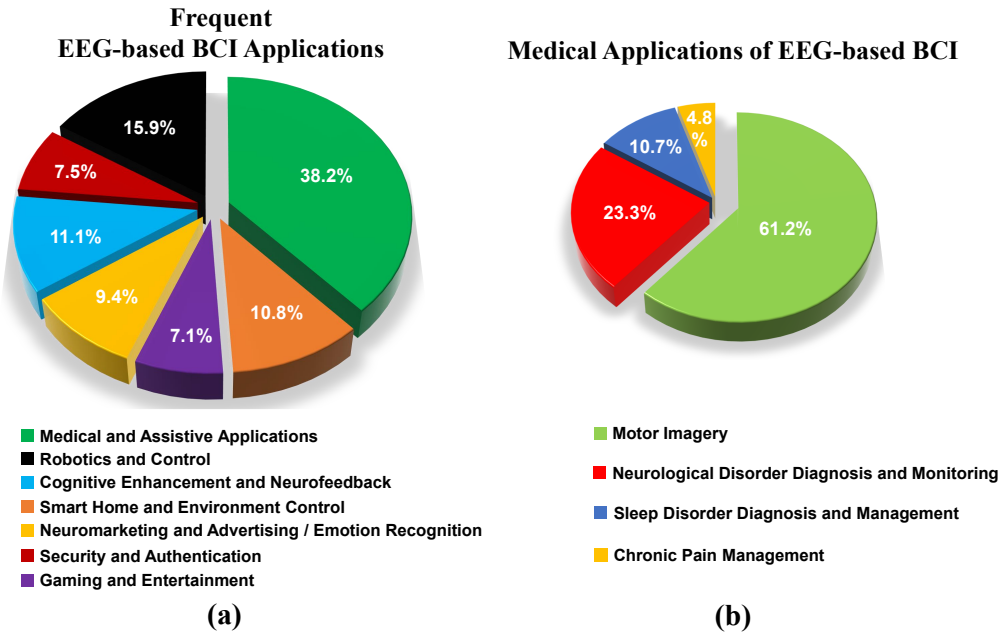

Fig. 30. Prevalence percentages of medical and non-medical applications of EEG-based BCI. (a) Distribution of major BCI application categories (N = 318 studies). (b) Distribution of medical EEG-based BCI applications (N = 122 studies). Percentages were calculated based on the studies included in the literature analysis described in Section II.

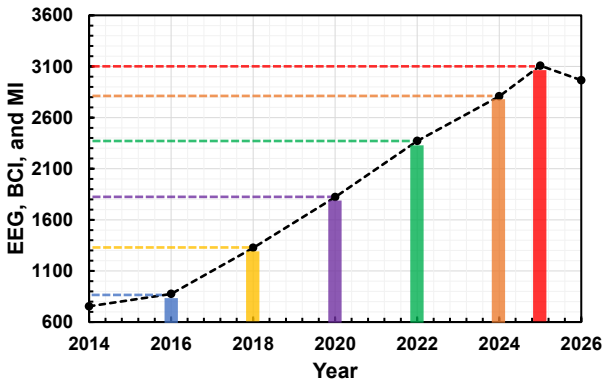

Fig. 31. Annual number of MI-BCI publications from January 2014 to July 2026, retrieved from Scopus using the search strategy described in Section II. The connecting line represents the yearly publication counts and does not indicate statistical regression, smoothing, or extrapolation.

movements while connected to EEG systems that provide real-time feedback. This approach not only aids in physical rehabilitation but also empowers users by restoring a sense of agency over their movements. Fig. 31 illustrates the increasing number of published works on MI-BCI applications in recent years. As research continues to advance in this field, the integration of AI algorithms and emerging technologies will further enhance the capabilities of MI-BCI systems, paving the way for innovative solutions in both clinical and everyday applications.

RQ2. What are the most frequently used signal processing techniques and datasets for EEG-based MI-BCI systems in the recent decade? Which processing approaches have demonstrated superior performance in recent studies and have the most potential to be used in future systems?

Based on the papers reviewed in Section V, several important observations can be made regarding the most frequently used signal processing techniques and datasets in EEG-based MI-BCI systems over the past decade. The analysis reveals the dominant algorithms employed in each major signal-processing block, including preprocessing, feature extraction, feature selection, and classification. It also highlights the datasets that have been most widely adopted for benchmarking and performance evaluation in recent studies. By comparing these trends with reported classification accuracies and implementation feasibility, it becomes possible to identify which processing approaches have demonstrated greater success in recent research and which ones show the strongest potential for integration into future EEG-based MI-BCI systems.

Among the signal preprocessing methods used in recent EEG based MI BCI studies, BWF is the most common approach, appearing in 30.71% of the reviewed works, as illustrated in Fig. 32 (a). It is followed by the NFL with 27.07%, which is primarily used for its effectiveness in suppressing power line interference. ICA accounts for 13.43% of the studies, while CAR follows with 10.81%, serving to improve spatial signal quality by reducing common noise across electrodes. The LAP filtering approach is utilized in 9.70% of cases, reflecting its role in enhancing local cortical activity relevant to MI tasks. Finally, DWT/CWT techniques appear in 8.28% of the studies, showing a more limited but still meaningful role in time–frequency based preprocessing.

Regarding feature extraction algorithm categories in recent EEG-based MI-BCI studies, spatial domain methods are the most frequently used, accounting for 23.20% of the reviewed approaches, as shown in Fig. 32 (b). They are followed by

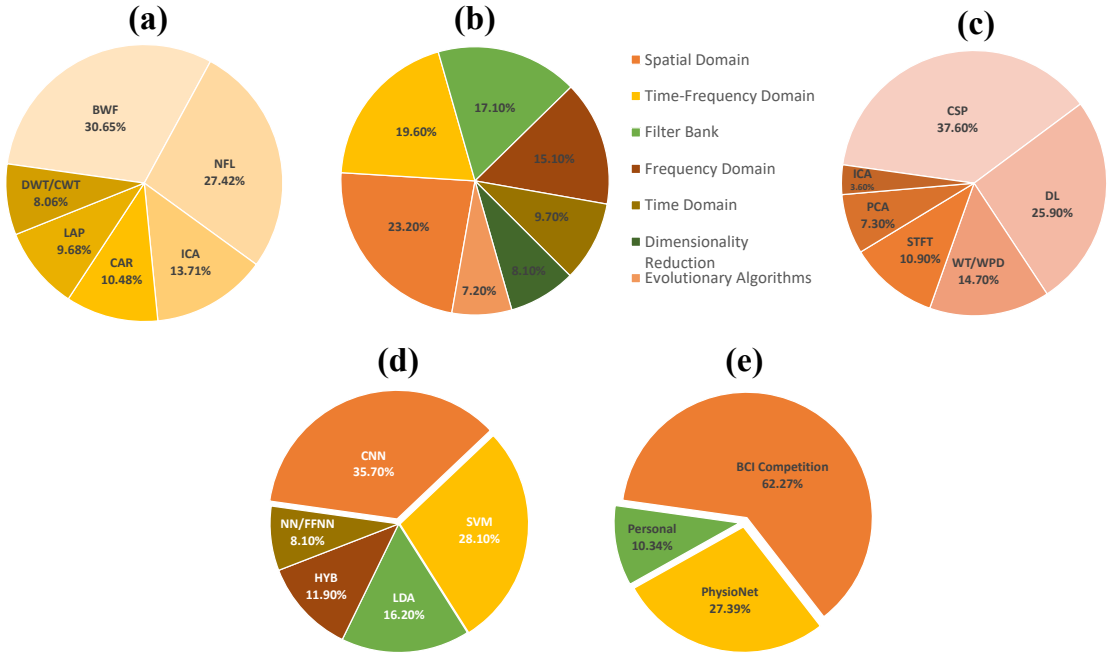

Fig. 32. Prevalence percentages of MI signal-processing algorithms and datasets based on the subsets of studies providing sufficient information for each analysis: (a) signal preprocessing algorithms ($N = 124$ studies); (b) feature-extraction algorithm categories ($N = 168$ studies); (c) feature-extraction algorithms ($N = 286$ studies); (d) feature-classification algorithms ($N = 297$ studies); and (e) common MI datasets ($N = 301$ studies).

time–frequency domain methods with 19.60%, highlighting the importance of capturing both temporal and spectral characteristics of EEG signals. Filter bank methods represent 17.10% of the studies, while frequency domain techniques account for 15.10%, reflecting their strong relevance for analyzing rhythm-based MI patterns. Time domain methods appear in 9.70% of the studies, showing a more limited but still meaningful role in direct signal analysis. Dimensionality reduction approaches account for 8.10%, indicating their usefulness in simplifying feature sets and reducing complexity. Finally, evolutionary algorithms are the least frequently used at 7.20%, suggesting a more specialized role in optimizing feature extraction processes.

Based on Fig. 32 (c), among the specific feature extraction algorithms used in EEG-based MI-BCI systems, CSP is the most dominant technique, accounting for 37.60% of the reviewed studies, mainly because of its strong ability to enhance discriminative spatial patterns between MI classes. LDA follows with 25.90%, reflecting its effectiveness in improving class separability and supporting robust feature discrimination. WT/WPD represents 14.70% of the studies, highlighting its usefulness in capturing the nonstationary time–frequency characteristics of EEG signals. STFT accounts for 10.90%, showing its continued relevance for analyzing spectral variations over time. PCA is used in 7.30% of the studies, indicating its role in reducing feature dimensionality while preserving important signal information. Finally, ICA appears in 3.60% of the reviewed works, suggesting a more limited but still meaningful application in artifact removal and feature enhancement.

Among the commonly used classification algorithms in EEG-based MI-BCI systems, CNN is the most prevalent, appearing in 35.70% of the reviewed studies, as depicted in Fig. 32 (d). This reflects the growing adoption of deep learning approaches capable of automatically learning spatial–temporal patterns from EEG data. SVM follows with 28.10%, due to its strong performance on high-dimensional and nonlinearly separable EEG features. Traditional classifiers remain relevant as well, with LDA used in 16.20% of studies, appreciated for its simplicity and fast computation. The hybrid (HYB) methods account for 11.90%, indicating increasing interest in combining multiple techniques to improve classification performance. Finally, NN/FFNN appears in 8.10% of the studies, showing that neural-based models continue to play a useful but less dominant role in MI-BCI classification.

Based on Fig. 32 (e), commonly used MI datasets in research include the BCI Competition datasets (62.27%), which are popular because they provide standardized benchmarks and a wide range of MI tasks; PhysioNet datasets (27.39%), which are appreciated for their open availability and detailed physiological recordings; and personal or experimental datasets (10.34%), which are usually gathered by individual research groups to support specific experimental designs or new protocols. Even when the same benchmark dataset is used, the classification setting can still differ considerably, ranging from a binary one-vs.-one task, such as left versus right hand, to one vs. rest tasks, such as imagined right-hand movement versus resting state, as well as multiclass problems involving three to five classes.

Direct comparison or averaging of classification accuracies across studies was not performed because the reported results were obtained using different datasets, numbers of classes,

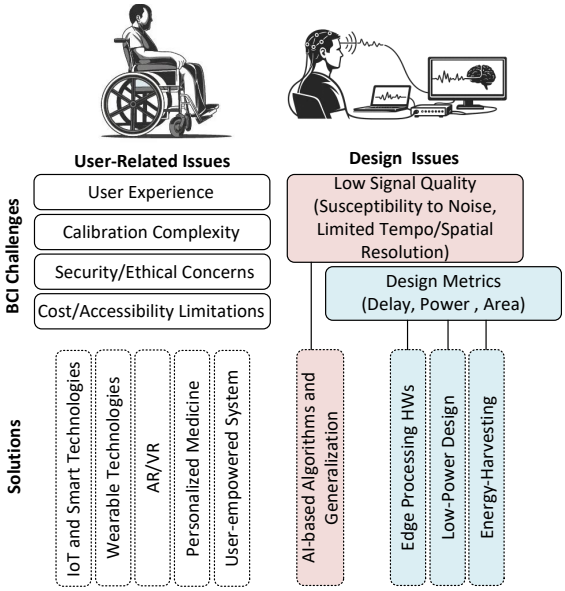

Fig. 33. Current MI-BCI systems challenges and solutions.

subject-dependent or subject-independent evaluation settings, and validation protocols. Accordingly, performance results are reported individually in Table IV and Table V, together with their corresponding experimental conditions.

RQ3. What are the challenges of EEG-based MI-BCI systems, and what are the promising candidate solutions?

The challenges and solutions associated with MI-BCI technologies can be grouped into two categories, user-oriented and design-oriented, as illustrated in Fig. 33. This classification provides a structured view of the diverse obstacles that hinder the practical deployment of MI-BCI systems and highlights the corresponding strategies proposed in the literature. Together, these categories help contextualize the multilayered nature of MI-BCI limitations and offer a unified framework to guide ongoing research and system development. The challenges and corresponding solutions summarized in Fig. 33 were synthesized from the reviewed literature and the studies discussed throughout Sections V and VI.

User-oriented challenges encompass those that directly affect usability and comfort, including discomfort from wet electrodes, bulky setups restricting mobility, and cognitive fatigue induced by sustained MI tasks. Long setup times and repeated calibrations further reduce accessibility, hindering adoption among non-expert users and patients [30]. Economic and ethical factors, such as high equipment costs and data privacy concerns, compound these barriers. Emerging solutions leverage IoT-enabled and wearable EEG systems, AR/VR integration [294], and personalized medicine to enhance user comfort and adaptability. Advances in affordable, low-power SoC architectures [271], [295]–[303], combined with cost efficient electrode materials such as dry or flexible sensors, significantly reduce production costs while improving comfort and usability [106]. Open-source ecosystems like OpenBCI, EEGNet, and BrainFlow facilitate lower software overhead, while cloud and edge computing employ lightweight ML and federated learning for real-time decoding on low-cost hardware. These user centric advancements, strengthened by IoT connectivity and smart environments, promote seamless interaction and everyday usability [188], [284], [304]–[316].

Design-oriented challenges relate to limitations in signal quality, system architecture, and real-time computational performance. EEG's weak and noise-prone signals, low spatial resolution, artifacts, and nonstationary inter-subject variability hinder accurate MI decoding, motivating advanced approaches such as DL, TL, Riemannian methods, and domain adaptation to learn more transferable representations and improve robustness across subjects and sessions [40], [131], [153], [159], [173], [177], [189], [191], [202], [317]–[320]. Implementation constraints such as latency, limited energy resources, and bulkiness are being addressed through EPoC, miniaturized wearables, hardware accelerators, and energy-efficient integrated-circuit techniques, including advanced-node FinFET technologies and low-power standard-cell architectures [321]. Additional improvements stem from advanced shielding against EMI/ESD [322], AI-enabled SoCs [323], ultra-low-power implants [324], and open-source SoC platforms like ESP4ML [325]. Future directions highlight neuromorphic computing, EH, sub-millisecond decoding, and quantum-inspired algorithms as key enablers of next-generation, real-time MI-BCI systems.

RQ4. How can various emerging technologies be brought together to create the next generation of EEG-based MI-BCI systems?

The trends in converging technologies for EEG-based BCIs from January 2014 to July 2026 are illustrated in Fig. 34 based on annual publication counts (the sources are listed in Section II), highlighting key developments in the field. The integration of AI (Fig. 34(a)) has enhanced feature extraction and classification, improving the accuracy of brain signal interpretation. The adoption of wearable technology (Fig. 34(b)) has made EEG-based BCIs more portable and comfortable, facilitating their use in daily activities and long-term monitoring. Hardware acceleration (Fig. 34(c)) has enabled power-efficient and high-performance processing, supporting real-time signal computation even in resource-constrained environments. The IoT (Fig. 34(d)) has enhanced BCI connectivity, enabling data exchange between devices for cloud-based analysis, remote healthcare, and adaptive neurofeedback applications. Edge processing (Fig. 34(e)) enables local EEG signal processing and MI decoding, potentially reducing communication latency, cloud dependence, and data transmission requirements while supporting privacy-aware and energy-efficient implementations. Finally, the convergence of EEG-based BCI and VR (Fig. 34(f)) has facilitated immersive neuroadaptive environments for neurorehabilitation, cognitive training, and human-computer interaction.

In the near future, the integration of emerging technologies will greatly improve the performance and accessibility of EEG-based MI-BCI systems. Advances in ultra-low-power processors, flexible bioelectronic interfaces, federated learning, and multimodal sensing will enable more adaptive and personalized BCIs. Enhanced wireless connectivity, cloud and edge AI, and smarter learning algorithms will deliver faster, more accurate, and intuitive brain-device interactions. As

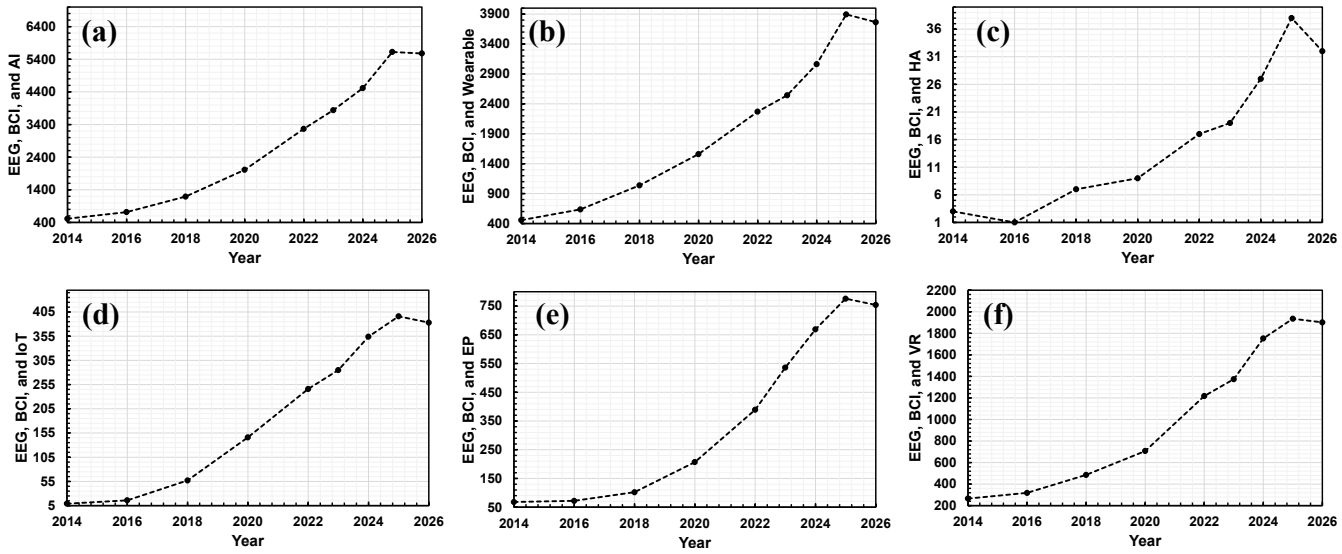

Fig. 34. Trends of converging technologies for EEG-based BCI from January 2014 to July 2026 based on annual publication counts retrieved from Scopus using the search strategy described in Section II: (a) EEG-based BCI and AI. (b) EEG-based BCI and wearable technology. (c) EEG-based BCI and HA technology. (d) EEG-based BCI and IoT. (e) EEG-based BCI and EP. (f) EEG-based BCI and VR. Scopus is used as the reference (see Section II for more details). The connecting lines represent yearly publication counts and do not indicate statistical regression, smoothing, or extrapolation.

BCIs become increasingly efficient and seamlessly integrated into digital ecosystems, they will expand applications across healthcare, rehabilitation, assistive devices, and consumer neurotechnology, bringing human-machine synergy closer than ever before.

IX. Conclusion

This paper has provided a comprehensive review of the technological evolution of EEG-based MI-BCI systems, synthesizing findings from a decade of research across software algorithms, hardware platforms, datasets, and application domains. By examining both statistical trends in the broader MI-BCI literature and detailed analyses of influential papers, the study highlights MI-BCI as the dominant and most impactful modality within non-invasive BCI research. The review demonstrates how advances in signal processing, feature extraction, and classification, especially those driven by modern AI models, have significantly enhanced the reliability and performance of MI-BCI systems for medical, assistive, and neurorehabilitation applications.

Despite substantial progress, EEG-based MI-BCIs continue to face challenges related to low SNR, limited spatial resolution, non-stationarity of neural signals, and the need for user-specific calibration. These limitations have historically restricted their accuracy, latency, and practicality in real-world environments. However, the field's trajectory shows that recent developments in deep learning, time-critical computation, and adaptive decoding algorithms are reshaping the landscape. These emerging methods are increasingly capable of handling complex EEG dynamics, reducing training time, and improving robustness across user sessions and environmental conditions.

On the hardware side, the convergence of SoC-based embedded processing, low-power accelerators, wearable EEG technologies, and IoT-enabled connectivity marks a significant shift toward portable and scalable MI-BCI architectures. Energy-efficient designs, wireless communication, and cloud/edge computing pipelines now permit real-time decoding and seamless integration with smart environments. Moreover, immersive VR ecosystems and neurofeedback-driven rehabilitation frameworks provide new avenues for enhancing user engagement, learning efficiency, and neuroplasticity, thereby widening the clinical and consumer applicability of MI-BCIs.

Looking forward, future research should focus on cost-effective and standardized MI-BCI platforms that unify AI-driven personalization, edge-level computation, and interoperable hardware architectures. The democratization of neurotechnology, supported by open-source toolchains, accessible datasets, and unified benchmarking protocols, will be essential for ensuring reproducibility and accelerating innovation. Ultimately, the continued convergence of AI, SoC design, IoT ecosystems, and immersive training technologies will shape the next generation of intelligent, adaptive, and practical EEG-based MI-BCI systems, enabling their transition from laboratory prototypes to widespread clinical, assistive, and consumer deployment.

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